LIST OF PARTICIPANTS #@APP-FORM-HERIAIA@#

Participant No.	Participant organisation name	Short Name	Country	Type of Organisation
1	Technische Universitaet Dresden	TUD	DE	University
2	HHLA SKY GmbH	HHLAS	DE	Industry
3	HFC Human-Factors-Consult GmbH	HFC	DE	Industry (SME)
4	AnyWi Technology BV	ANYWI	NL	Industry (SME)
5	Technolution BV	TCN	NL	Industry (SME)
6	Beyond Vision	BYV	PT	Industry (SME)
7	Innovation Disco	IVD	GR	Industry (SME)

LIST OF ABBREVIATIONS

Abbreviation	Complete designation	Abbreviation	Complete designation
ALTAI	Assessment List for Trustworthy Artificial Intelligence	ICAO	International Civil Aviation Organization
AMC	Acceptable Means of Compliance	KPI	Key Performance Indicator
ATM	Air Traffic Management	LTE	Long-Term Evolution (4G mobile communication standard
BVLOS	Beyond Visual Line of Sight	QoS	Quality of Service
B2B	Business-to-Business	RF	Radio Frequency
C2	Command and Control	ROI	Return on Investment
DAR	Dynamic Airspace Reconfiguration	SESAR JU	Single European Sky- Joint Undertaking
DMP	Data Management Plan	SSH	Social Sciences and Humanities
DPIA	Data Protection Impact Assessment	SWOT	Strengths, Weaknesses, Opportunities, Threats *(analysis method)*
dronnect	Drone Connectivity and Traffic Management (project acronym)	TG	Target Group
EC	European Commission	UAS	Unmanned Aerial System (drones)
EU	European Union	USSP	U-Space Service Provider
EASA	European Union Aviation Safety Agency	U-Space	Uncrewed airspace Traffic Management
EUROCAE	European Organisation for Civil Aviation Equipment	UTM	Unmanned Traffic Management
GDPR	General Data Protection Regulation	VLL	Very Low-Level airspace
GEP	Gender Equality Plan	VR	Virtual Reality
GM	Guidance Material	WP	Work Package(s)
HMI	Human-Machine Interface	5G	Fifth Generation mobile communication standard

1. EXCELLENCE #@REL-EVA-RE@#

dronnect is our proposal to establish a resilient European drone ecosystem: one that creates **green jobs1**, develops **new skills**[1], strengthens **industrial competitiveness**, and supports both the **European Green Deal**[2] and the **Cities Mission**. Our interdisciplinary consortium integrates **technology, regulation, and societal acceptance** into a coherent methodology. Our consortium brings together leading experts across engineering, social sciences, humanities, policy, and business innovation, ensuring that every aspect of drone deployment is examined through multiple lenses. *dronnect* does not only demonstrate what is technically possible – it makes large-scale drone logistics **technologically feasible, socially desirable, economically viable, and environmentally sustainable**.

Over the past decade, drones have evolved from niche technologies into indispensable assets. Recent global events have highlighted how unmanned aerial systems are becoming **ubiquitous** and **central,** not only to **European security and sovereignty** but also to the **resilience and competitiveness** of the broader European economy. At the same time, drones are enabling rapidly expanding civilian markets: they support **precision agriculture**, accelerate **infrastructure inspection**, **disaster response**, and **environmental monitoring**, while laying the foundations for **urban air mobility** and **automated logistics**. This strategic and economic relevance underscores the urgency of building a resilient and competitive European drone ecosystem. The **Drone Strategy 2.0**[3] provides the framework for this ambition. Foundational enablers such as the **EASA IAM Hub**[4] and **U-space concepts**[5] have laid the groundwork – but their true value can only be realised through **large-scale, real-world implementation**.

Yet, a critical gap remains: **public trust and reliable performance**. Citizens, authorities, and potential business users ask: *Can drones operate safely at scale? How will contingency landings be managed in dense urban environments? Can drone services guarantee availability, resilience, and ROI compared to road-based logistics?* If drones are perceived as unreliable, adoption will stall – regardless of technological promise. This uncertainty is further compounded by the risk of misuse by malevolent actors. This is precisely the bottleneck the dronnect consortium addresses. **Rotterdam and Hamburg**, with their critical infrastructure of ports, warehouses, and peri-urban zones, provide real-world pilot environments to test **resilience, contingency landing procedures, operational safety, and public acceptance**. Insights from these demonstrations will directly inform **scalable and replicable deployment pathways.** Building on the state-of-the-art advances and EU flagship initiatives, dronnect transforms identified gaps in **feasibility, governance, and societal acceptance** into tangible solutions. Our approach is **interdisciplinary and human-centered**, combining **engineering innovation, social sciences and humanities (SSH) expertise, and governance approaches**. Importantly, **everything is open**: the project commits to **open science, open access, open data, and open-source tools**, ensuring transparency, reusability, and broad uptake of results.

dronnect stands on **three integrated pillars**:

- **Technological reliability and safety** – advancing resilient communication infrastructures, digital tools, smart traffic management systems, and **contingency landing site protocols** to guarantee safe, consistent BVLOS operations.
- **Societal trust-building** – SSH experts go beyond surveys, employing immersive VR-based experiments on **noise, safety perception, and sensory acceptance**, engaging citizens in live demonstrations, and fostering participatory dialogue. This approach achieves **positive behavioral acceptance**[6], directly addressing concerns and bridging trust.
- **Economic sustainability and ROI** – co-created **B2B business models** with stakeholders demonstrate how drone logistics can reduce downtime, improve performance, and open new markets. This ensures drone-based logistics is not only greener but also **reliable and cost-effective** compared to current road-based transport.

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[1] New skills: The project proposes solutions to train remote operators to handle multiple drones simultaneously.

[2] The European Green Deal https://commission.europa.eu/strategy-and-policy/priorities-2019-2024/european-green-deal_en

[3] EU Drone Strategy 2.0 https://transport.ec.europa.eu/system/files/2022-11/COM_2022_652_drone_strategy_2.0.pdf

[4] EASA IAM Innovation Hub https://www.easa.europa.eu/en/domains/drones-air-mobility/drones-air-mobility-landscape/innovative-air-mobility-hub

[5] EASA U-Space Concept EU 2021/664 https://www.easa.europa.eu/en/document-library/easy-access-rules/easy-access-rules-u-space-regulation-eu-2021664

[6] Enhancements made based on EASA report https://www.easa.europa.eu/sites/default/files/dfu/uam-full-report.pdf

1.1. OBJECTIVES AND AMBITION #@PRJ-OBJ-PO@#

dronnect aims to **build and demonstrate a resilient, safe and secure communication infrastructure** that ensures consistent and reliable information exchange between **the digital traffic management platforms of remote operators**[7] **and the proposed digital online platform of multi-level institutional systems**[8], required for operational coordination, cooperation, and the fair use of VLL U-Space airspace. Through **targeted demonstrations and experimental surveys**[9], the project will actively engage the general public, building awareness of drone technologies, addressing societal concerns, and fostering acceptance for the sustainable integration of Beyond Visual Line of Sight (BVLOS)[10] drone operations in both urban and peri-urban environments. At the **local level**, *dronnect* is **designed to pioneer and support EASA and SESAR 3**[11] **initiatives on U-space services and IAM concepts**, showcasing adaptive innovation that is responsive, efficient, and characterized by minimal downtime within an innovative, fast, and sustainable B2B drone logistics model. At the **European level**, the project will strengthen Europe's global leadership, ensuring that the deployment of drone services reflects European values of safety, inclusiveness, and sustainability [12] while enhancing competitiveness and sovereignty in the rapidly evolving drone sector.

TECHNICAL AND SOCIETAL OBJECTIVES

dronnect builds on its societal and governance foundations to **design, implement, and validate a resilient, multi-link communication backbone for traffic management and coordination between remote operators, civil aviation authorities, and municipalities, ensuring mission continuity under coverage gaps, interference, security threats, or network degradation**. Rather than relying on a single technology, the project will validate an integrated and redundant architecture, supported by real-time performance monitoring to enable rapid contingency actions and reinforce operator trust. Embedded within U-space and UTM services, this backbone will guarantee uninterrupted situational awareness, reliable data exchange, and low-latency responsiveness, conflict resolutions, or emergency contingencies, or emergency contingencies, while **ensuring seamless interoperability across VLL corridors, U-Spaces, CTRs, non-U-space environments, and cross-border operations in line with the EU Drone Strategy 2.0.** Recognizing that technical robustness alone is insufficient for adoption, *dronnect* will **research and quantify public acceptance thresholds through surveys, immersive scenario testing, and case studies** across European cities, while **co-creating digital governance models that empower municipalities to authorize, co-supervise, and coordinate VLL operations with their jurisdiction**, including contingency planning with operators and USSPs. Dedicated communication strategies will **transparently inform citizens on drone operations and emergency measures**, thereby addressing the societal and governance dimensions critical to trust, acceptance, and large-scale deployment of drone services. In alignment with emerging standards for digital unmanned aviation services, dronnect tackles the higher-level research challenges essential for enabling future EU-wide deployment:

Technical Research Questions	Societal and Governance Research Questions
• *What supervisory ratios for remote operators can safely be defined under U-space and ICAO[13] constraints?* • *How can multi-link redundancy architectures enhance resilience at scale, ensuring operational continuity under regional outages or disruptions?* • *Which harmonized protocols, service interfaces, and interoperability frameworks are required to guarantee safe traffic flow and priority management between U-space and non-U-space airspace across Member States?* • *What impact will ad-hoc airspace reconfigurations by ATC have on active U-space users?*	• *How do citizens' acceptance thresholds vary across use cases and socio-demographic contexts?* • *What communication and transparency mechanisms most effectively increase public trust during contingency scenarios?* • *How can municipalities act as operational partners to U-space service providers in managing dense VLL traffic safely and fairly?* • *What viable B2B business models can underpin sustainable drone logistics services, and how can responsibilities, costs, and revenues be shared*

[7] **Remote operator reference to the person in the control center commanding the BVLOS operations**
[8] EU Digital Transition https://commission.europa.eu/strategy-and-policy/priorities-2019-2024/europe-fit-digital-age_en
[9] Experimental surveys are VR based surveys to identify people's sensitivity to drone noise, and other sensory impacts
[10] BVLOS operations in specific category https://www.easa.europa.eu/en/domains/drones-air-mobility/operating-drone/specific-category-civil-drones
[11] SESAR U-Space initiative https://www.sesarju.eu/U-space
[12] EU Sustainable & Smart Mobility initiative part of EU Green Deal https://transport.ec.europa.eu/transport-themes/eu-mobility-transport-achievements-2019-2024/sustainable-smart-mobility_en
[13] ICAO UTM core principles https://www.icao.int/sites/default/files/left-menu-pdfs/UTM%20Framework%20Edition%204.pdf

• *How can communication performance metrics feed into European regulatory frameworks[14],[15], supporting certification and standardization efforts?*	*between operators, municipalities, and service providers?*

Against this backdrop, ***dronnect* defines five strategic technical and societal objectives**, each directly addressing the research questions above and operationalized through the activities in methodology, pilot actions, and section 3.

#O1 SCIENTIFIC AND TECHNOLOGICAL OBJECTIVE

Design and demonstrate resilient and reliable connectivity for drone BVLOS operations in urban, peri-urban environments.

- **Develop and integrate multi-link communication backbone** (5G, LTE, RF, satellite) into drone control systems for BVLOS operations in line with U-space regulatory requirements.
- The KPIs for communication infrastructure not well qualified because, currently there is no adequate definition for the QoS needed for these systems. **Define QoS metrics and protocols** for resilient communication over commercial networks.
- **Implement dashboards** and advanced failover/load-balancing logic to ensure operational continuity under network degradation.
- **Demonstrate and validate communication algorithms**, system performance, and onboard resource optimization through test setups.

No.	Key Performance Indicator	Target
1.1	Availability and continuity of the communication link	Above 99%
1.2	Low latency	Less than 50 milliseconds
1.3	Resource utilization for information exchange	Few kilobits per second
1.4	Network performance based contingency action	Success rate above 85%

#O2 SCIENTIFIC AND TECHNOLOGICAL OBJECTIVE

Develop and implement a resilient, interoperable management framework that enables seamless integration of U-Spaces, and ATM for urban and peri-urban BVLOS drone operations.

- **Identify and define operational requirements** by analysing challenges of connecting U-space and non-U-space BVLOS logistics operations, including preparation of take-off and landing sites, and assessing the impact of disruptions (e.g., Dynamic Airspace Restrictions issued by ATM).
- **Design, develop, and validate workflows, strategies, and guidelines** for seamless entry, exit, and traversal of U-space during BVLOS missions, including contingency procedures for simultaneous landings of multiple UAS under DAR conditions.
- **Develop and evaluate enabling infrastructure and HMI solutions**, focusing on take-off/landing site readiness, remote pilot supervisory capacity (one operator managing multiple drones), and integration with U-space services to ensure safe and scalable fleet operations.
- **Plan, prepare, and demonstrate two pilot scenarios[16]** drone logistics operations, and an emergency response demonstrator, to validate technical performance, safety, and regulatory alignment under real-world urban and peri-urban conditions.
- **Conduct cross-demonstration analysis** and extract lessons learned, developing strategies, operational guidelines, and systemization pathways to support EU-wide deployment and standardization of BVLOS drone logistics in U-space environments

No	Key performance Indicator	Target
2.1	HMI Interface – Feasibility of remote-operator managing multiple drones simultaneously	5 to 8 drones per remote pilot
2.2	Performance of safe contingency landings due to temporary airspace restrictions or DAR of U-Space airspace	80% success at pre-assigned safety landing points 20% safe landing at dynamically assigned points

[14] EASA Working Groups for RPAS standards https://www.easa.europa.eu/en/downloads/137252/en
[15] EUROCAE Working Group for C3 & S https://www.eurocae.net/working-group/wg-105-sg-2/
[16] **Pilot demonstration is a scaled-down, preliminary test of a new technology, process, or product in a controlled environment that closely mimics real-world conditions**

#O3 SCIENTIFIC AND TECHNOLOGICAL OBJECTIVE

Design and demonstrate a collaborative, digital platform enabling coordinated multi-level governance interaction— to facilitate operational governance, share best practices, and support the societal readiness for drone integration.

- **Identify policy gaps and develop governance models** for the integration of drones into urban and peri-urban ecosystems, ensuring alignment with municipal regulations, societal expectations, and sustainable city planning objectives.
- **Design, develop, and implement** a digital application platform that actively involves municipal authorities in drone traffic management, while analysing and defining strategies for the seamless integration of drone logistics into the broader **multi-modal urban transportation ecosystem**.

No	Key Performance Indicator	Target
3.1	Digital platform for municipal authorities to enforce contingence landings	90% successful with remote operator coordination
3.2	Digital platform used by municipal authorities	3+ municipal authorities used and provide feedback
3.3	Coordinate with multi-modal transport (ground, water)	80% successful integration trough 2 pilot demonstration

#O4 SOCIETAL AND ECONOMIC OBJECTIVE

Assess responsiveness and adapt innovative measures to strengthen societal readiness and public acceptance of long-distance drone operations, enabling sustainable logistics services

- **Identify and quantify key public concerns** and acceptance factors by conducting large-scale surveys on societal readiness for drone operations in urban and peri-urban contexts.
- **Design and deploy immersive VR-based scenario testing** to evaluate public reactions to different operational concepts, noise levels, transparency measures, and contingency situations.
- **Integrate findings into an expanded European Drone Acceptance Framework**, providing evidence-based insights that link societal expectations with technical, regulatory, and operational requirements.
- **Develop and validate targeted mitigation strategies** to foster public trust, reduce resistance, and support the responsible deployment of drone logistics and emergency services in diverse urban environments.
- **Collaborate with the Support Action HORIZON-CL5-2026-01-D2-09** to obtain feedback, align methodologies, and reinforce policy recommendations through coordinated European-wide actions.

No	Key Performance Indicator	Target
4.1	Address public concerns and sentiments for positive scaling of acceptance for VLL U-Space airspaces	80% positive with urban and peri-urban public
4.2	Developed mitigation solutions with a focus on public interest	50% Reduction in concern rates measured in VR experimental study

#05 SOCIETAL AND ECONOMIC OBJECTIVE

Design and validate a fast, sustainable, low-downtime B2B drone logistics model that strengthens user, stakeholder confidence, supports economic growth in EU drone logistics market

- **Design and validate a fast, sustainable, and low-downtime B2B drone logistics model** by conducting early-phase stakeholder questionnaires, use case analyses, and requirements mapping to capture operator, customer, and regulator expectations. This will ensure that KPIs are aligned with operational needs and market opportunities from the outset.
- **Develop and test strategies, tools, and deployment pathways** for scalable B2B drone logistics services, including business model evaluation, ROI opportunities, early adopters benefit stakeholder engagement, and governance alignment, in order to strengthen user and stakeholder confidence while supporting economic growth and competitiveness of the EU drone logistics market.

No	Key Performance Indicator	Target
5.1	ROI and early adopters benefit in drone logistics market	80% success rate across various stakeholders

5.2	B2B model for fast pace rollout of logistics drone operation	50% above success rate for cities like Hamburg and Rotterdam

AMBITION

The following section outlines the foundational EU frameworks and flagship projects on IAM and U-space, while highlighting the critical gaps that dronnect goes beyond by addressing through an integrated, process-driven approach.

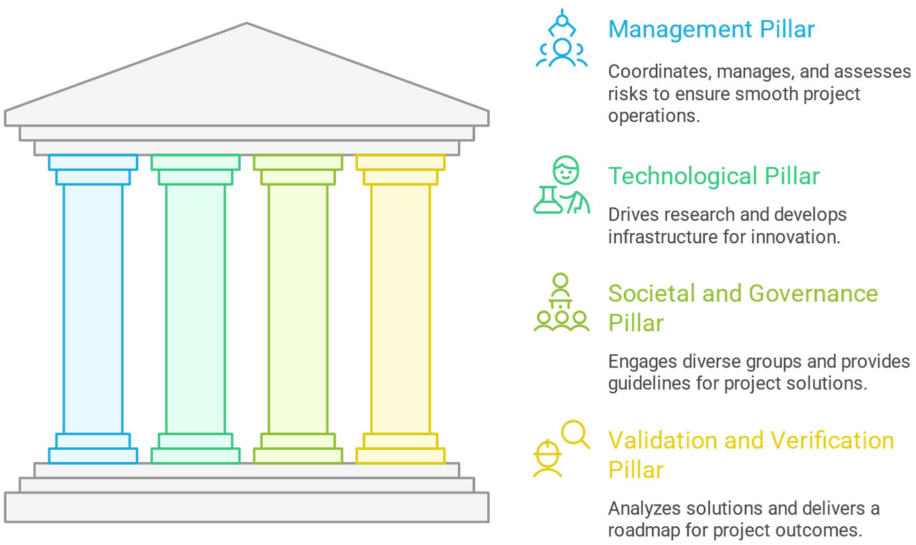
Dimension	State of Art (EU projects and initiatives)	Gaps and Limitation	R &I Progress beyond
Reliable Communication	SESAR JU GOF2.0 (2020), DronC2om(2020), 5G!Drones (2020), EALU-AER (2025), ADACORSA (2023)	Mostly lab-scale prototypes, single-link focus. No unified multi-link architecture validated with U-space involving various elements of C2 integration. Lack of failover protocols.	*dronnect* designs and validates multi-link architectures (5G/LTE, RF, Non-terrestrial network/satellite). Defines UAS-specific protocols and QoS metrics, implements resilient failover methods, and provides a network management dashboard that ensures compliance with U-space/EASA requirements. **Attain TRL 5**
UTM & U-space Operational Infrastructure	PODIUM (2020), ENSURE (2026), EuroDRONE (2022)	Fragmented demos; little to now understanding of seamless transitions between U-space and non-U-space BVLOS ops. Contingency workflows (e.g., DAR) underexplored. Take-off/landing site readiness in mass contingency not systematically validated.	*dronnect* develops end-to-end workflows for U-space transitions, validates DAR contingency ops, and delivers HMI operator and design guidelines. Conducts drone logistics & contingency demonstrators in urban/peri-urban environments. **Attain TRL 5**
Societal Acceptance & Human Factors	AiRMOUR(2023) RECIPROCITY (2023), EASA study (2022)	Limited scale & diversity of surveys; weak linkage between societal findings and technical design. Little testing in emergency/disruption contexts. The sensory fears and sensitivity is sparsely addressed	*dronnect* executes EU-wide surveys, immersive VR-based testing (noise, transparency, DAR/emergencies). Creates an expanded EU Drone Acceptance Framework, directly feeding into system design & policy guidelines-
Tools for Municipal Governance & Policy	ADACORSA (2023), Flying Forward (2020)	Municipalities rarely active in U-space operations. No digital governance tools linking cities to U-space providers. Weak integration with multimodal mobility planning.	*dronnect* develops beyond planning and integrate the municipal governance via digital platform for active involvement in U-space. Designs integration strategies into multimodal ecosystems. Provides policy gap analysis + governance models aligned with city planning.
B2B Market Models & Deployment Pathways	ADACORSA (2023), AiRMOUR(2023)	Remain at feasibility stage, lacking validated, scalable, low-downtime B2B logistics models. Early stakeholder needs to be systematically mapped.	*dronnect* delivers validated B2B logistics models (fast, low-downtime). Conducts stakeholder-driven KPI mapping early. Provides tools, strategies, deployment pathways for rapid scale-up of EU drone logistics markets.

Table 1.1: Summary of Project Ambition #§PRJ-OBJ-P

1.2 Methodology #@CON-MET-CM@# #@COM-PLE-CP@#

dronnect adopts an **integrated, multi-disciplinary framework structured around four major pillars as visualised in figure 1,** each directly aligned with one of the project's core objectives. These pillars correspond to dedicated work packages (WPs) and draw on the expertise and resources of a diverse, pan-European consortium encompassing telecommunications, aviation, societal research, and business model exploitation specialists. Active alignment and structured collaboration with stakeholders, municipal authorities, and European UAM initiatives will bridge industrial needs with project ambitions. The consortium partners are deeply embedded in networks with local, port, and municipal authorities, ensuring direct alignment between project outcomes and real-world deployment environments[17]. This co-creation approach ensures that the technical, regulatory, societal, and business pillars (figure 1.1) are not developed in isolation, but operate as an interconnected framework. By fostering early engagement and continuous dialogue with business actors and end-users, the project will establish a shared direction that maximises impact, accelerates adoption, and reinforces the coherence of all activities.

Figure 1.1: Representation of dronnect pillars #§PRJ-OBJ-PO§#



The **management pillar (WP1-5)** coordinates, manages, assesses risks, and supports activities carried out throughout the project's lifespan. It encompasses communication, public outreach, engagement with external industrial providers and end-users, as well as the organization of public demonstrations through communication, dissemination, and exploitation activities. The **technological pillar (WP6-8)** drives fundamental research, the development of operational infrastructure, a reliable communication architecture, and corresponding platforms that support research and innovation while delivering tangible results. The **societal and governance pillar (WP9-11)** plays a key role in reaching diverse target groups and enabling project partners to evaluate their solutions, thereby providing relevant guidelines for the work programme. Finally, the **validation and verification pillar (WP 12-15)** are critical in analysing the solutions and delivering a roadmap that advances project results into both short- and long-term outcomes and impacts, which are further detailed in Section 2. The description of each pillar directly translates into corresponding work packages and activities presented in Section 3. Together, these interconnected pillars form the structural foundation of the project, enabling a systematic progression through its main stages. It spans from preparatory research and development through to real-world demonstration, with the ultimate aim of reaching **Technology Readiness Level (TRL) 5** for all critical components and workflows by project end.

- **Concept Definition and Preparation:**

The project begins with a comprehensive phase focused on eliciting, consolidating, and formalizing the technical, operational, and industrial requirements necessary for large-scale, BVLOS drone logistics in VLL airspace. This includes mapping the needs of all relevant stakeholders (operators, regulators, municipalities, end users), analysing the integration challenges posed by U-space and non-U-space operations, and defining high-level system and API architectures. *Contributors: All partners*

[17] **TCN's** Mobimaestro **platform (https://www.technolution.com/move/mobimaestro/) manages all traffic lights and digital traffic systems in Rotterdam, the Port of Rotterdam, and the Province of South-Holland, integrating with parking enforcement, emergency services, and public transport. In addition, TCN collaborates with the Port of Rotterdam in the Horizon** MODI **project (https://modiproject.eu/) on autonomous truck transport, covering secure communication, localisation, and mission-critical safety. Although not a formal partner, the Port actively supports our implementation at a Rotterdam container terminal.**

- **Requirement Analysis and Design:**

Building on the requirements (WP 5-8), the consortium will develop interoperable, resilient multi-link communication systems, new communications network management algorithms, and aviation-grade QoS metrics. In parallel, exchange protocols for seamless U-space and non-U-space integration, workflow optimizations for UAS operators, and HMI solutions supporting scalable remote operations will be engineered and validated in laboratory/testbed environments. *Contributors: telecommunication, UTM, and UAV partners*

- **Prototype Development and Simulation**

Parallel to technical innovation, the project conducts large scale surveys, immersive VR scenario testing, and structured engagement workshops to capture and quantify societal readiness and acceptance factors. Policy gap analyses and governance model blueprints are developed in collaboration with municipalities, ensuring societal needs and regulatory frameworks (WP 9-11) are embedded in technical solutions. *Contributors: technology partners, Social science, policy, city partners*

- **Integrated Pilot Demonstrations - Operational Validation and SSH feedback:**

Advanced prototypes are field-deployed and demonstrated in two major operational pilots (WP 12,13)—one focused on logistics and one on emergency response—in representative urban and peri-urban environments. These tests validate the end-to-end system under real-world conditions, with special attention to handover between U-space and non-U-space, simultaneous emergency contingency handling of multiple drones, operator workload, and B2B logistics readiness. *Contributors: All partners, local authorities, U-space providers*

- **Impact Evaluation, Exploitation, and Roadmap:**

A systematic cross-analysis of all demonstrations is carried out to extract lessons learned, refine technical and operational guidelines, and optimize deployment and governance pathways. Continuous feedback loops with the Support Action HORIZON-CL5-2026-01-D2-09, further stakeholder engagement, and business model evaluation are conducted to support EU-wide upscaling and standardization. *Contributors: Business, policy, dissemination partners*

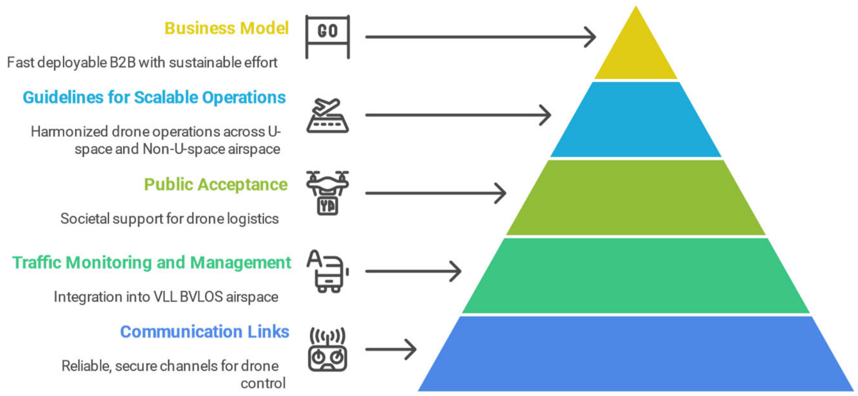
The project's methods are directly aligned with its technical, operational, and societal objectives, addressing key research questions through the deliverables and key milestones in section 3. Aforementioned they deliver an extended European Drone Acceptance Framework that integrates societal insights, regulatory compliance, and municipal governance. Simultaneously, on technical aspect - validated reliable and resilient communication architectures, failover mechanisms, and reference protocols with USSPs will enable scalable, safe, and U-space-compliant remote operations of drone fleets, fully aligned with EASA regulations and SESAR IAM objectives, ensuring interoperability and harmonization across European airspace.

TECHNICAL CONCEPT, ASSUMPTIONS, AND CURRENT GAPS

To achieve an economically viable and scalable urban drone logistics ecosystem, a robust and interoperable framework of supporting services is required—one that ensures high availability, safety, and security in line with EASA and SESAR IAM objectives. The most pressing enabler is the establishment of **reliable, resilient, and secure communication links** that support highly automated monitoring, real-time human oversight, and reliable intervention options (as in figure 1.2). Without dependable command, control, and telemetry channels[18] particularly under challenging urban conditions such as signal degradation, multipath effects, and frequent handovers across mobile networks—large-scale deployment cannot be realized.

[18] RPAS critical communication link - https://www.easa.europa.eu/sites/default/files/dfu/SC-RPAS.C2-01_RPAS%20Command%20and%20Control%20-%20proposal.pdf

Figure 1.2: Methods in dronnect – pyramid for drone logistics services deployment #§PRJ-OBJ-PO§#



A second critical prerequisite is the **integration of UAV operations into multimodal traffic management systems,** notably U-space and ATM/UTM platforms. While U-space provides a cooperative system for managing drone operations, many practical challenges still have to be solved to achieve operational status. These include: the operational and technical complexity of entering, exiting, and crossing U-spaces; the lack of well-defined procedures and status communication to all stakeholders when U-space airspace is reclaimed by ATC (DAR); how many drones can swiftly divert to dynamically allocatable safe landing sites in contingency situations; and public concerns regarding noise, privacy, and routing near residential areas. Furthermore, although U-space is designed to incorporate smart data sources from diverse stakeholders, its current scope does not fully address intelligent flight scheduling, prioritization of e.g., medical flights, adaptive routing, or integration with broader smart-city infrastructures. Building on this, a key challenge lies in enabling seamless handovers between systems, operators, and communication links during drone missions. To ensure high availability and resilience, communication design must function as a robust supporting layer—capable of handling mobility-driven transitions, interoperability between networks, and unexpected interference—while guaranteeing continuous operational safety. The project goes beyond demonstrating technical feasibility by embedding robustness, failover mechanisms, and resilience directly into the workflow, procedures, and communication system design—thus laying the foundation for large-scale UAV deployment and public trust.
The project's methods are directly aligned with its technical, operational, and societal objectives, addressing key research questions through the deliverables and milestones described in Section 3. Specifically, the work will deliver:

- An **extended European Drone Acceptance Framework**, incorporating societal insights, regulatory alignment, and municipal governance.
- **Validated communication architectures** with redundancy, failover, and reference protocols interoperable with USSPs.
- **Scalable, U-space–compliant operations** aligned with SESAR IAM objectives, ensuring harmonization across European airspace.

Yet, even with robust technical and regulatory enablers, the long-term viability of drone logistics also depends on societal acceptance, without which deployment will face resistance and delays.

PUBLIC ACCEPTANCE AS AN ENABLER FOR RESPONSIBLE DRONE DEPLOYMENT

Currently, public acceptance is one of the most underdeveloped critical preconditions for the successful deployment of drone-based logistics services. While regulatory initiatives and technical pilots are advancing, the majority of ongoing projects treat societal acceptance as a secondary or reactive task, frequently restricted to general public opinion surveys**. There is a lack of context-sensitive, use-case-oriented research that provides decision-makers with insights into public concerns, perceived risks, and acceptance thresholds.** The AiRMOUR project offers valuable lessons for understanding societal readiness, particularly by highlighting the importance of context in public acceptance, the role of familiarity and exposure in building trust, the need to address concerns such as social inequality, and the effectiveness of live demonstrations combined with participatory engagement methods. Building on these validated insights, this project proposes a methodologically rigorous, empirically grounded, and forward-looking framework that goes beyond generic acceptance studies by integrating context-sensitive approaches, immersive simulations, and a two-path empirical strategy to capture both general readiness and situational acceptance, thereby enabling municipalities, operators, users of drone services, and policymakers to proactively address societal barriers and design inclusive, widely supported drone logistics solutions.

- a **drone acceptance framework** tailored to logistics operations in urban environments,
- an **immersive VR-based method** to simulate realistic drone scenarios and measure public responses, and
- a **two-path empirical strategy** that captures both general readiness and situational acceptance.

Enabling municipalities, operators, and policymakers to foresee societal barriers prior to deployment creates **more inclusive solutions**. Compared to the current state of the art, this approach connects abstract acceptance models with tangible, scenario-based feedback and effective B2B model (as mentioned in Section 2).

PILOT ACTIONS

Figure 1.3: AI generated illustration of drone logistics remote piloting #§PRJ-OBJ-PO§#

The *dronnect* project will validate its technological, operational, and societal innovations through **two pilot demonstrations** in **Hamburg and Rotterdam with established intent of support and cooperation**. Both cities represent highly relevant operational environments: they combine **critical logistics infrastructure (ports, warehouses, multimodal hubs)** with **dense urban districts**, and they are committed to ambitious climate neutrality targets (Hamburg: **40% CO_2 reduction by 2030, climate neutrality by 2050**[19]; Rotterdam: **Smart City and Zero-Emission Zone goals**)[20]. These conditions create **complex, safety-critical Very Low-Level (VLL) airspace operations (figure 1.4, 1.7)**, making them ideal testbeds for validating **U-Space integration, resilient communication, and societal acceptance measures**. Also, the trial demonstrations are not only technical validations but also highly visible public showcases of correct system behaviour under all circumstances. Their credibility depends on flawless performance: every handover, contingency landing, and coordination workflow must operate reliably and transparently. These demonstrations are therefore critical, as they serve both to validate the robustness of the technology and to build trust among citizens, authorities, and stakeholders in the safety and resilience of large-scale drone operations.

The demonstrations will be conducted in Months 30–33, following extensive technical validation and regulatory approval.

Pilot 1: Hamburg – Logistics Demonstration

The main objective of this demonstration is to

- Validate reliable drone logistics (represented figure 1.3) operations in and out of U-space between port warehouses and urban districts, in complex, advanced scenarios, with the use of advanced U-space services.
- Demonstrate UAS-operator tools, procedures, and workflows involved in entering/ exiting U-space and in contingency situations
- Showcase multi-link resilient communications (5G/LTE + satellite fallback) integrated with U-space services.

[19] https://www.hamburg.com/visitors/getting-around/moinzukunft-22640

[20] https://research-and-innovation.ec.europa.eu/system/files/2021-09/cities_mission_implementation_plan.pdf

- Engage enterprise stakeholders and the general public to showcase efficiency and benefits of drone logistics, increase awareness, and gather feedback or concerns.
- Demonstrate real-time network performance monitoring dashboard (figure 1.7) for safe and resilient drone operations.

Figure 1.4: Complex VLL airspace – Hamburg #§PRJ-OBJ-PO§#

HHLAS logistics drones(in figure 1.5) will operate along a dedicated imaginary VLL corridor between a Point A and Point B. Operations will be monitored by a **remote-control centre (as shown in figure 1.6)** connected to a **U-Space Service Provider (USSP) and (simulated) ATC tools**. The flights will also test handover resilience across multiple communication links and visualize performance via a network dashboard.

Figure 1.5: HHLAS "market-ready" drone#§PRJ-OBJ-PO§#

Figure 1.6: HHLAS "market-ready" remote operator centre#§PRJ-OBJ-PO§#

The key goal of this pilot action is not only to validate technology but also to create a shared learning moment through a **"Dronnect Demo Day: Shaping the Future of Urban Drone Logistics"**, where stakeholders, users of drones services, and the general public will actively engage in dialogue.

The demonstration pilot includes a **round-table discussion**, bringing together city authorities, logistics operators, drone service providers, and civil society representatives. Topics will include:

- *How drones can safely integrate into city logistics without disrupting everyday life and work.*
- *The role of complex **U-Space services and resilient communications** in ensuring reliable and scalable operations.*
- *The potential of drone logistics to **support sustainable urban mobility goals** (e.g., reducing congestion, emissions).*
- *Public concerns around **noise, safety, privacy, and equitable access**.*
- *Feedback on the **Drone Acceptance Framework**, informed by live demonstrations and interactive tools.*

Figure 1.7: AI based illustration of network management dashboard #§PRJ-OBJ-PO§#

To capture societal sentiment, **SSH-led surveys and interactive engagement methods** (including immersive VR tools) will be deployed on-site, ensuring that citizen voices directly feed into the project roadmap and guidelines. **Effectiveness of this demonstration** focus on validating the safe and resilient performance of next-level U-Space procedures and workflows for UAS-operators in complex scenarios, and demonstrating robust communication handovers amongst connected systems and processes. Equally important are societal KPIs, including the effective engagement of stakeholders, the depth of public dialogue generated, and the degree to which participants leave with a stronger understanding of and confidence in drone-based logistics solutions.

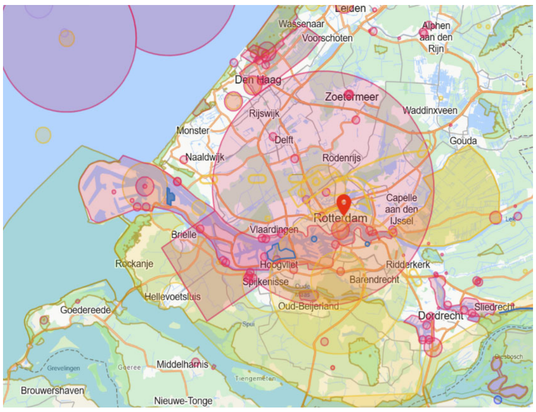
Tools involved in the demonstration	Description
HHLAS drones	Market-ready drones for logistics operations (used in WP 12)
HHLAS Integrated Control Centre	Market-ready, web-based flight planning, management, monitoring, and control platform (used in WP 12)
HHLAS UTM platform	Visualization of routes, no-fly zones/ DAR and complex integration of external and conformant U-space services (WP 7,12)
Network monitoring dashboard	**Supporting the remote operator for effective C2 performance Outcome of project (TRL 5) (WP 6, 12)**
Dronnect demo day - public survey	**Outcome through exploitation of the project (WP 15)**
Stakeholder roundtable	**Identify the business, societal, governance challenges. Outcome through exploitation (WP 14,15)**

Pilot 2: Rotterdam – Integration of Municipal Governance & Emergency Integration

Traditionally, the absence of designated contingency take-off and landing sites in Very Low Level (VLL) corridors poses significant risks during emergency situations—such as sudden corridor closures—since drones in flight may lack safe, pre-approved locations to reroute and land, thus jeopardizing operational reliability, safety, and regulatory compliance for all stakeholders involved. Building upon this critical gap, the object of this demonstration is

- ***Demonstrate dynamic contingency management capabilities** (communicating geo-fencing, emergency landing sites) in an urban environment.*
- *Showcase **real-time authority–operator coordination** capabilities via digital platforms.*

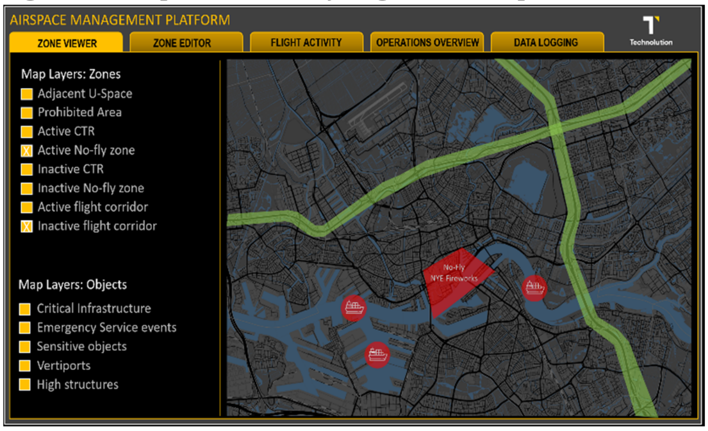
Figure 1.7: Complex VLL airspace – Rotterdam #§PRJ-OBJ-PO§#



A staged urban emergency (temporary VLL airspace closure due to a ground incident) will require drones in flight to reroute. The remote operator will receive dynamic geo-fencing updates and coordinate with municipal officials via the network-managed UTM platform. Drones will be tasked to land safely at pre-identified contingency sites.

Figure1.8: Beyond Vision remote operator platform and "in-market" drone #§PRJ-OBJ-PO§#

Figure1.9: Representation of Digital Municipal Governance Platform#§PRJ-OBJ-PO§#



The **effectiveness of the pilot** will be measured by the successful implementation of dynamic geo-fencing updates, the ability to coordinate safe contingency landings during emergencies, rapid authority–operator communication (using platforms figure 1.9, 1.10) validation of complete governance workflows under realistic scenarios, active engagement of multiple municipal departments in live demonstrations, and the development of actionable policy recommendations based on the pilot outcomes.

Tools involved in the demonstration	**Description**
Beyond Vision drone	In market – quadcopter drone with capabilities for logistic operations (Used in WP 13)
Map & Operation Platform	BeXStream – "Market ready" remote operator platform for visualization and contingency routing (Used in WP 13)
Digital Municipal Governance Platform	**Real-time monitoring Platform of VLL operations, authority–operator coordination** **Outcome of project(TRL 5) (WP 11, 13)**
Discussion with local and critical infrastructure authorities	**Potential use of platform, and specific requirements are received and a roadmap for integration provided through exploitation (WP 11, 15)**

Table 1.3 : Summary of tools (used and outcomes) of dronnect – Rotterdam Pilot

RELEVANT NATIONAL OR INTERNATIONAL RESEARCH AND INNOVATION ACTIVITIES AND LINKS TO DRONNECT

dronnect is deeply embedded in relevant national and international research and innovation ecosystems. In Rotterdam, the consortium builds on support as a TCN- member of **droneport Rotterdam** the municipality, the Port of Rotterdam, and the Province of South-Holland, with integration into parking enforcement, emergency services, and public transport. This strong foundation is extended through collaboration in the Horizon Europe MODI project, focused on autonomous truck transport in the port area, addressing mission-critical communication, localisation, and safety. In Hamburg, dronnect benefits from the active support of the Port Authority and the City of Hamburg, while HHLAS—an industrial-scale drone operations provider is a full consortium partner. The project is further connected to ongoing European initiatives such as BLU-Space, CIVITAS, UIC committee activities, AiRMour, UNCHAIN, Reciprocity, CITYAM, Windrove and CodeZero.

These links ensure that dronnect outcomes are not developed in isolation but co-shaped with complementary projects, and through the **stakeholder round-table feedback** (D12.1)- allowing cross-fertilisation of results, replication in new cities, and reinforcement of EU-wide objectives for safe, sustainable, and trusted drone ecosystems

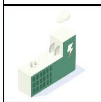
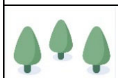
INTERDISCIPLINARY APPROACH

*dronnec*t brings together a truly interdisciplinary consortium spanning **engineering, natural sciences, and social sciences** to address the technological, operational, and societal dimensions of urban drone logistics. Under **Engineering and Technology**, partners contribute expertise in drone and autonomous aviation systems, secure communications, multimodal integration, traffic management, and software platforms - forming the core of the project's **technological pillar**. Within the **Natural and Information Sciences**, the focus lies on advanced data management, monitoring, data validation, and verification, enabling the assessment of demonstrative solutions and ensuring the safe integration of logistics drones into VLL airspace. **Social Sciences and Humanities (SSH)** are at the heart of *dronnect*, ensuring *human-centered design and operational safety* through applied ergonomics and human–machine interaction research. The **societal and governance pillar** builds on this by driving citizen engagement, communication, and exploitation activities to strengthen public acceptance and ensure alignment with regulatory and policy frameworks. SSH contributions specifically address: **Economic aspects** – supporting market uptake, investment planning, technology transfer, and replication in the B2B landscape. **Education and communication** – raising awareness among local stakeholders and end users on the benefits of drone-based mobility and logistics through training, outreach, and dissemination. **Societal aspects** – engaging citizens in co-design processes, evaluating co-benefits, and addressing public concerns and uncertainties to build trust and social acceptance. **Political science** – enabling capacity building and upskilling of public authorities and policymakers, supported by digital platforms for governance and decision-making.

1.3. GUIDELINES AND ADHERENCE TO EU POLICIES

DO NOT SIGNIFICANT HARM

dronnect strictly adheres to the "Do No Significant Harm" (DNSH) principle of the EU Taxonomy, ensuring that all research, development, and demonstration, commercialisation activities contribute positively without causing adverse environmental impacts. Through innovative methodologies and continuous monitoring, the project aligns fully with the six environmental objectives:

Environmental Objective	Description
 climate change mitigation	The project adheres to the principles of climate change mitigation, with drone logistics operations directly contributing to the reduction of carbon emissions and **supporting the carbon neutrality goals of the pilot cities of Hamburg, Dresden & Rotterdam.**
 climate change adaptation	Our approach ensures no harm to climate change adaptation efforts, as resilient system **design and operational planning are aligned with long-term adaptation needs.**
 sustainable use & protection of water & marine resources	By reducing the reliance on ground vehicles and enabling sustainable logistics solutions in urban and peri-urban ports, the project fully complies with relevant requirements and standards while ensuring **no adverse impact on water and marine resources.**
 transition to a circular economy	The project aims to develop a sustainable and viable lifecycle for the solution- business models for drone logistics that optimise resource utilisation while fully respecting the principles of the circular economy, with a **strong focus on reusability and recyclability.**
 pollution prevention & control	The project supports **pollution prevention by replacing carbon-intensive deliveries with electric drones.** It also protects biodiversity and ecosystem health by reducing urban noise, minimizing ground traffic in green spaces, and carefully planning flight paths to avoid sensitive areas.
 protection and restoration of biodiversity & ecosystems	**All pilot demonstrations will be conducted in compliance with local environmental regulations**, and continuous monitoring will ensure adherence to DNSH principles throughout the project lifecycle.

ADDRESSING SCOPE OF THE CALL

The description of each pillar directly translates into corresponding work packages and activities that addresses the scope of the call presented in Section 3.

Scope of the European Call	Pillar of *dronnect*	Level	Description
Prepare a roadmap new/upgraded infrastructure and logistics	Technological, Societal and Governance	Fully addressed	Technical assessment and specific regulatory requirements such as resilient communication backbone, operational requirement of logistic corridors and traffic management and coordination
Guidelines for integrating drone infrastructure (vertiports, landing areas, charging stations)	Societal and Governance, Verification and Validation	Fully addressed	The planed round table discussion addresses the challenges of VLL drone operations and its impact in viable B2B model
Assess effects of IAM traffic and vertiport requirements	Technological, Verification and Validation	Supportive	The drones used in the pilot demonstration are adhering to EASA standards
Noise monitoring needs and tools	Technological	Supportive	Adapts flight planning based on noise sensitive zones (such as hospitals)
Integration with existing freight flows and multimodality	Technological, Verification and Validation	Fully addressed	Evaluates potential interfaces between UTM systems and logistics IT platforms, and multiple scenarios operation alongside of ground robots in hospitals, warehouses
Demonstration of IAM cargo delivery in cities	Technological, Societal and Governance, Verification and Validation	Fully addressed	Two key pilot demonstrations planned to validate and verify innovative solutions developed for *dronnect*
Impact and benefits assessment (congestion, noise, pollution, airspace capacity)	Societal and Governance	Fully addressed	Assessment from survey and discussion, collaboration from stakeholder, logistics operators, city authorities and general public - live demonstration and immersive VR based study
Development of viable business models for sustainable IAM logistics	Verification and Validation, Societal and Governance	Fully addressed	Design fast and less downtime deployment B2B framework, investment strategies, EU cities replication roadmaps
Public awareness and citizen buy-in	Verification and Validation, Societal and Governance	Partially addressed	Leads citizen engagement, immersive VR-based acceptance studies. Qualitative advancement benefits - ROI for early adopters
Lessons learnt, replication, and training packages	All Pillars	Fully addressed	Contributes technical inputs, guidelines, competence-building materials, and replication strategies for European cities through communication, dissemination and exploitation activities.
Rebound effects and sufficiency analysis (energy, large-scale ops)	Validation and Verification	Partially addressed	Explores technical and societal dimensions of rebound effects and sufficiency, including equity, scalable limits of drone operations in VLL and sustainability.

Fully addressed: Complete societal and/or technical assessment carried out
Partially addressed: A selected aspects of societal and/or technical assessment
Supportive: Indirectly addressed as outcome of another activity

GENDER DIMENSIONS IN DRONNECT R&I ACTIVITIES

The drone sector remains women significantly underrepresented in technical, operational, and decision-making roles. *dronnect* addresses gender equality as a cross-cutting principle in all stages of R&I, ranging from team composition and stakeholder engagement to technology design and societal impact analysis.
Gender-equal participation in team: *dronnect* is committed to balanced gender representation in all levels of project participations, targeting 50/50 gender balance in leadership, technical teams, and work package coordination, supported by institutional Gender Equality Plans (GEPs). **Inclusive engagement:** *dronnect* promotes inclusive engagement and dissemination with balanced stakeholder participation and gender-sensitive communication. All public communication follows gender-sensitive language and design principles. **SSH action:** *dronnect* WP 9, 10 collects and analyses gender-disaggregated data with intersectional dimensions to evaluate potential differences in how drones affect perceptions on mobility, trust, safety, and acceptance. The public survey includes gender as an analytical dimension. VR-based scenario testing incorporates gender analysis to identify potential differential impacts (needs, concerns, consideration, responsiveness, satisfaction) of drone operations on men and women. **Feedback and mitigation strategies:** Findings from gender-sensitive analysis will be fed back into the development and mitigation strategies, ensuring drone operations are inclusive and responsive. This includes recommendations for campaigns and operational transparency.
This concise approach ensures *dronnect*'s full compliance with Horizon Europe's gender equality goals while enhancing the quality and societal relevance of research and innovation.

OPEN SCIENCE PRACTICES

*dronnec*t embraces comprehensive Open Science practices aligned with the European Commission's commitment to open, collaborative research.
Science communication: All scientific publication will be available under open access conditions. *dronnect*'s scientific outcomes will be openly accessible (primary or secondary publications) through trusted repository such as **Zenodo, Open Research Europe**, **ArXiv**, **Qucosa**, **OpenAire** for public deliverables. Exceptions apply only to internal, sensitive data between the partners in the project. The available resources with data, metadata will be available in project's website and other informed via other communication channels. To facilitate these commitments, project beneficiaries have allocated a budget for publication. To benefit the scientific community and enable collaboration and advancements in the software and solutions, open-source protocols, and tools will be used and transparently shared through public repositories. Also, the project's Data management plan and action as a part of WP 1, 2 will be built on the principles of **Research Data Management (RDM) in line with FAIR (Findability, Accessibility, Interoperability, Reusability)** principles. **Science education resources:** The **communication and dissemination actions** (WP 3,4) will enable open knowledge transfer and activities ensure research findings reach diverse audiences beyond academia, supporting informed public discourse on innovative air mobility technologies. **Demonstration and workshops at schools, universities as part of the project's actions** will put forth the advantages and the positive impact (potential new jobs and skills, sustainable logistic mobility) to integrate the community. **Citizen engagement:** The project actively promotes **citizen science engagement** via public demonstrations, stakeholder co-creation workshops, and transparent pilot results reporting.

DATA AND OTHER RESOURCES MANAGEMENT

This project will generate and manage diverse types of essential data such as experimental communication performance metrics, flight telemetry data, network traffic logs, survey responses, VR experimental results, demonstration video recordings, reports and documents. All personal data collected (e.g., from surveys) and its use by SSH expertise in the project will be handled in full compliance with the **EU (Regulation 2018/679)** General Data Protection Regulation (GDPR). As forementioned in section 1.2.4.4, the Data Management Plan (DMP), developed as an integral part of the project, adheres to FAIR principles and ensures responsible handling of research data throughout the project lifecycle. The DMP outputs will be machine-readable (XML), machine-actionable (JSON), and exportable to PDF format via the **OpenAIRE ARGOS** service. Project-related research data and outputs will be made openly accessible by default, unless justified restrictions are required (e.g., to protect personal data, intellectual property, or sensitive information). Privacy protection will be addressed with particular rigor through both organizational and technological safeguards, including secure data storage solutions, federated learning approaches for privacy-preserving analytics, advanced data encryption, and access control mechanisms. A Data Protection Impact Assessment (DPIA) will be conducted to identify and mitigate potential risks. The project commits to ensuring that data is: **Findable** – each dataset will be assigned a unique name and a persistent identifier; if deposited in a trusted repository, it will be accompanied by comprehensive metadata to support discovery and reuse. **Accessible** – scientific and technical results will be shared openly whenever possible; when access restrictions are necessary, data owners will provide clear justification (e.g., for GDPR compliance, IP protection, or commercial agreements). **Interoperable** – datasets will be stored in standardized, widely supported formats, compatible with free and open-source software applications. **Reusable** – data will be made available under well-defined access conditions, ranging from open access to restricted or embargoed access, as determined by the consortium. Each partner will be responsible for implementing the DMP within their work packages and ensuring compliance with FAIR principles for the data they generate and process. For long-term storage and preservation, data will be deposited in trusted public repositories, such as **Open Research Europe**, ensuring sustainability, accessibility, and security of research outputs.

SUMMARY OF THE SECTION

- The project's rigorous integration of technical and public acceptance strategies offers a future where drones serve society responsibly, making cities smarter, safer, sustainable mobility.
- *dronnect*'s participatory approach co-creates innovative B2B models and governance platforms, nurturing confidence, new skills, and green jobs for resilient European growth.
- dronnect bridges the trust gap between authorities, stakeholders, and citizens, actively listening to public concerns and fostering more positive acceptance through immersive VR engagement and inclusive demonstrations.
- The project ignites hope for a new era of mobility, advancing resilient drone traffic management in urban and peri-urban environments especially with critical infrastructure like ports to unlock economic and societal benefits across Very Low Level (VLL) U-Space airspace.

#§CON-MET-CM§# #§COM-PLE-CP§# #§REL-EVA-RE§#

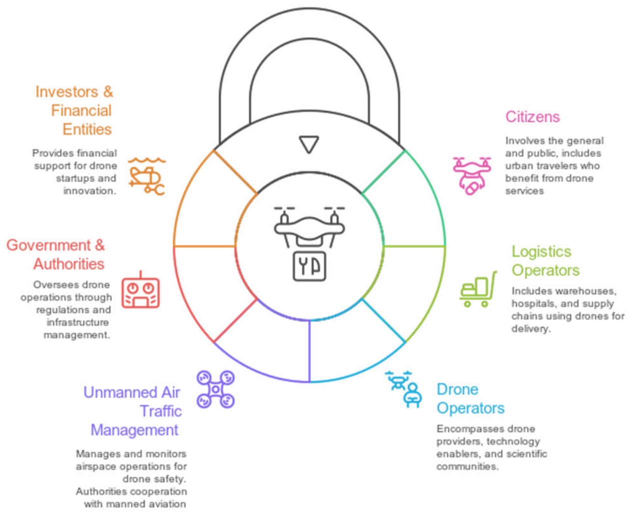
2. IMPACT #@IMP-ACT-IA@#

2.1. PROJECT'S PATHWAYS TOWARDS IMPACT

dronnect will deliver five major results, each directly aligned with the expected outcomes of the call:

- ☑ Resilient Multi-Link Communication Infrastructure for reliable and resilient drone operations.
- ☑ Validated Operational Framework for seamless U-space integration and scalable fleet management.
- ☑ Municipal Digital Platform for democratic drone traffic governance and integration into urban mobility systems.
- ☑ Public Acceptance Framework with evidence-based mitigation strategies for societal readiness.
- ☑ Economically Viable B2B Drone Logistics Models to strengthen EU competitiveness and market uptake.

The objectives are perceived into respective pillars and based on the deliverables of associated WPs as described in section 3, these results are designed to create tangible, measurable improvements for the key stakeholder groups, with tailored impacts and KPIs. The project address to 6 major classification of target groups



- **Citizens:** This group involve uninvolved personnel. The subgroups classified with general public and directly involved communities – like people in urban. People travelling to urban, peri-urban locations via other modes of transport
- **Logistics Operators:** Logistic operators, institutions like warehouses, hospitals, labs, supply chains or other offices who use the drone logistics for last-mile, last-meters service
- **Drone Operators:** This group has sub-group of the actual drone provider for logistic operators and their associated service like remote fleet management operator, technology enablers like communication, navigation, battery, drone manufactures and scientific communities working on key principles of the technology who are bound to satisfy EASA, EUROCAE standardisation and provide services based on SLAs.

Figure 2.1 : Unlocking potential of drone logistics – Target group mapping

- **Air traffic Management Providers:** This includes who are managing, monitoring and cooperating the operations in the airspace like ATM/ANSP, UTM, USSP, CISP.
- **Government & Authorities:** This group overarches municipalities, critical infrastructure authorities, and regional city authorities. Also, includes standardisation bodies and European regulatory agencies (EASA, SESAR JU).
- **External Investors & Financial Entities:** This group involved indirect people but economically bound entities like venture capital funds and private investors in drone/UAM start-ups, corporate investors from logistics, telecom, and mobility sectors, public funding agencies and innovation support instruments, insurance and risk management entities supporting drone services.

Through **coordinated communication, dissemination, and demonstration activities**, *dronnect* ensures that each result reaches the relevant stakeholder group, maximising both short-term adoption and long-term transformation of Europe's drone ecosystem.

2.2. EXPECTED OUTCOME TO EXPECTED IMPACT

EO (Expected Outcome) and EI (Expected Impact) are expected outcome of the call, and long-term impact of the dronnect at EU level. The proposed action (from objectives in section 1 and the activities in section 3) builds a coherent pathway from concrete **technological, operational, and societal outcomes (EO1–EO5)** to the broader **policy and market impacts (EI1–EI3)**, ensuring full alignment with the Drone Strategy 2.0 and the EU's climate-neutral, digital transition objectives.

SHORT-TERM EXPECTED OUTCOMES

The 5 main results of the project on a shorter term deliver the following expected outcomes as discussed below

#	EO title	Description	Pillar & WP	Verification (Deliverable)	Impact Target Group
1	Resilient and Reliable Communication Infrastructure	Reliable communication infrastructure, establishing key communication links to remote operators, USSP, and municipal digital platforms. The work evaluates protocols, handover strategies, and KPIs (bandwidth, acceptable latency, error and loss rates)	Technological, verification and validation pillars (WP 5,6,7,8, 15)	D 6.1, D6.2 with DEM - D12.2, D15.3	1.Drone Operators: Enhanced operational reliability through >99.9% communication availability and <50ms API response times 2.ATM Providers: Standardized interfaces enabling seamless integration with existing UTM/U-space systems 3.Government & Authorities: Technical compliance frameworks supporting EASA certification and regulatory harmonization
2	Operational Framework for Seamless U-Space Integration	Development of comprehensive operational workflows that ensure safe and seamless transitions between U-space and non-U-space environments in urban and peri-urban drone logistics. Establishment of operational infrastructure and procedures for take-off/landing sites, disruption management (e.g., dynamic airspace restrictions), and simultaneous multi-UAS operations under emergency conditions. Validation of the operational framework in real-world settings, including public demonstrations and emergency response exercises in coordination with city authorities.	Technological, societal and Governance, verification and validation pillar (WP 5,7,8,11,12, W15)	D7.1, D7.2, D8.1, D8.2, D12.2, D15.1, D15.2	1.Drone Operators: Operational certainty through >98% emergency procedure success rates, standardized workflows, and validated infrastructure readiness protocols for take-off/landing sites 2.ATM Providers: Clear integration protocols reducing coordination complexity and operational risks, with proven strategies for managing Dynamic Airspace Restrictions and simultaneous UAS operations 3.Government & Authorities: Validated compliance procedures supporting regulatory approval and cross-border harmonization, including infrastructure standards and emergency response coordination protocols
3	Municipal Governance & Digital Platform	The project develops digital tools and governance frameworks that empower local authorities to actively participate in drone traffic management and integrate drone services into urban mobility systems. These solutions are validated through real-world demonstrations, ensuring functionality, stakeholder engagement, and alignment with municipal decision-making processes.	Societal and Governance, verification and validation pillars (WP 5,9,10,11,13 15)	D 9.1, D9.2, D11.1, D15.5	1.Government & Authorities: Enhanced democratic oversight through digital platforms deployed in 3+ cities with >80% municipal engagement rates 2.Citizens: Increased transparency and trust through accessible monitoring systems and participatory governance mechanisms 3.Logistics Operators: Streamlined permit processes and operational coordination through integrated municipal interfaces
4	Drone Acceptance Framework and mitigation strategies	The project builds societal readiness by establishing baseline acceptance metrics and conducting large-scale surveys to capture citizen perspectives across Europe. It advances this through VR-based experimental studies, enabling immersive testing of public reactions and the co-	Societal and Governance, validation and verification pillars (WP 5,9,10,11,12,15)	D 9.1, D9.2, D12.2, D15.5	1.Citizens: Evidence-based engagement strategies achieving certain improvement in acceptance rates across diverse urban contexts 2.Government & Authorities: Policy recommendations and mitigation frameworks supporting responsible deployment, informed by VR-validated public reactions

		design of mitigation strategies. The results converge into a validated framework for societal acceptance, providing practical interventions to foster trust and support adoption of drone logistics.			3.External Investors: Risk reduction through validated social acceptance models and mitigation strategies, enhancing investment confidence in large-scale drone services
5	Economic Competitiveness and Market Growth	Sustainable business models and deployment strategies that align with stakeholder expectations and unlock market opportunities for drone logistics in Europe. By combining market analysis with large-scale demonstrations, it provides a scalable roadmap for B2B operations, ensuring economic viability and reinforcing EU leadership in innovative air mobility.	Technological, Societal and Governance, validation and verification pillars (WP 5-14,15)	D5.1, D15.4	1.Logistics Operators: Validated B2B models with clear ROI calculations and risk assessments across 5+ logistics use cases 2.External Investors: Market-ready deployment strategies that in turn supports the grand vision of the €14.5 billion drone services market target by 2030 3.Drone Operators: Scalable service frameworks enabling transition from pilot projects to commercial operations

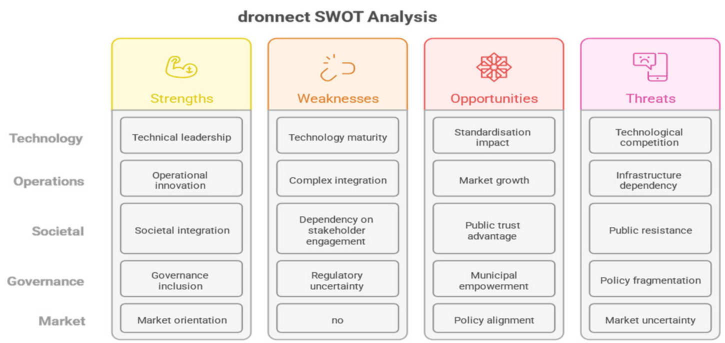
The above mentioned EO address the expected outcomes of the call as addressed in table below

Call Expected Outcome	**Contribution from Project**	**Level of Achievement (Indicator)**
Enabling very low-level unmanned aviation (IAM) for sustainable and smart urban mobility in cities, aligned with Drone Strategy 2.0	EO1 (Resilient communication backbone), EO2 (Operational framework for BVLOS/U-space), O5 (Market growth)	**High** – validated through 2 real-world demonstrations and technical KPIs supporting Drone Strategy 2.0
Institutional capacities to enable IAM are built up	EO3 (Municipal governance & digital platform), EO4 (Acceptance framework), O5 (Training packages via WP14)	**Medium–High** – digital platforms tested in ≥3 cities, capacity-building guidelines for authorities
New tools and services for optimising IAM in cities and workable governance arrangements	EO1 (Communication architecture), EO2 (Operational infrastructure and workflows), EO3 (Digital governance platform)	**High** – TRL5 validated prototypes, governance platform pilots, standardized U-Space interfaces
Evidence-based guidelines and recommendations for cities on developing a sustainable UAM ecosystem	EO2 (Infrastructure standards),EO3 (Municipal integration), O4 (Societal acceptance guidelines), O5 (Business roadmaps)	**High** – Roadmaps (D14.2, D14.3) plus regulatory recommendations from pilots and VR studies
Advanced understanding and quantification of the value of IAM, particularly for logistics	EO2 (Operational performance metrics), EO4 (Societal benefit quantification), EO5 (Economic ROI/business models)	**High** – Evidence generated via pilot demonstrations, surveys, ROI analysis
Enhanced multimodality, urban logistics planning/flow and communication between stakeholders	EO1 (Multi-link resilient comms), EO2 (Integration with existing freight flows), EO3 (Municipal authority collaboration), EO5 (B2B stakeholder engagement)	**Medium–High** – Demonstrated in port/urban logistics corridors in Hamburg & Rotterdam
Creation of jobs and economic growth from UAM services in the long term	EO5 (Validated B2B business models, scaling strategy), EO4 (Societal trust building reducing adoption barriers)	**Medium** – Business model roadmap (2030 horizon), direct exploitation via consortium members
Responsiveness to diverse social needs, increasing beneficial societal uptake and trust	EO4 (Acceptance framework + VR testing), EO3 (Citizen-facing transparency tools), O5 (Stakeholder incentives for early adopters)	**High** – Evidence-based acceptance strategies, participatory governance, and mitigation tools

Taken together, the consortium covers the full innovation (as mentioned in below SWOT analysis and table) chain to create expected impact towards advancing the EU drone economy.

Partner	Key expertise	Connection to external stakeholders
TU Dresden (TUD)	Advanced robotics, HMI, 5G/6G communications architecture and protocols	Interfaces with Telekom providers and IAM research communities
HHLA SKY (HHLAS)	Technology solution provider for advanced drone operations & fleet management, Certified UAS operator, UTM integration, industrial-scale drone logistics solutions	Established links to drone logistics operators, ports, city authorities, regulators, ministries, aviation stakeholders, industry partners
Human Factors-Consult (HFC)	Societal Acceptance & Ergonomics, VR Lab, user acceptance studies,	Connects to citizens, stakeholders, training bodies
AnyWi Technologies (ANYWI)	Secure software architectures, API development, multi-service integration	Established links to drone operators, platform developers, logistics users
Technolution (TCN)	Traffic & Infrastructure Management	Stakeholder links to municipalities, mobility-as-a-service, multimodal transport systems
Beyond Vision (BYV)	BVLOS operational expertise, UTM software, UAV system integration	Connects to drone operators, emergency services, regulators
Innovation Disco (IVD)	Expertise in business model, communication, dissemination, and exploitation of research and innovation projects	Connects to municipalities, UAM/IAM networks, policy makers, citizen communities

Table: Summary of consortium expertise and established collaboration with external stakeholder for *dronnect*



MID-TERM OUTCOME TO IMPACT PATHWAY

Through extended reflection on the project outcomes, a mid-term evaluation can track the progress towards long-term impact

- >95% communication reliability across multi-link networks; 15–30% reduction in mission downtime; 15–20% lower drone deployment costs; adoption in at least **5 EU cities** supporting Drone Strategy 2.0
- ≥3 municipal digital governance pilots operational; ≥200 city staff trained; >40% increase in citizen trust (survey-based); 10–15% reduction in municipal oversight costs
- TRL6 prototypes validated; interoperable U-space interfaces integrated; at least **3 new digital services for IAM** tested with city partners
- Publication of **regulatory roadmaps** and **3 sets of city-level guidelines**; policy recommendations integrated into ≥2 municipal mobility strategies; citizen engagement tested with VR/AR tools
- Demonstrated **50% faster last-mile deliveries**; CO_2 emissions reduced by ≥25% per delivery vs. road; ROI analysis showing 3–5x returns for adopters; quantified benefits published in ≥3 scientific/industry reports
- Demonstrated integration of drones into **2 multimodal logistics corridors** (e.g., ports, urban hubs); 20% efficiency gain in freight flows; establishment of **public-private stakeholder platforms** in 3 cities
- **1,000+ new skilled jobs training and created** in early adopter regions; 40% market uptake projected within 5 years; validated B2B models replicated across ≥3 EU cities

EXPECTED IMPACT

The large scale expected impact of the outcomes are represented as heatmap that align the KPIs and benefits

EI1 – Accelerating the EU Drone Strategy 2.0

By delivering a resilient, multi-link communication backbone and seamless U-space integration workflows, dronnect will enable safe, scalable, and efficient drone operations across Europe. This will reduce operational delays by up to 30%, lower communication failures to below 0.1% outage rates, and support the EU Drone Strategy 2.0 vision of a pan-European drone services market. The interoperable backbone is expected to cut average drone deployment costs by 15–20%, enabling faster commercial adoption.

KPIs & Economic Benefits:

- >95% communication reliability in multi-link operations.
- 20–30% reduction in mission downtime due to communication disruptions.
- 40% projected market uptake potential within 5 years, supporting growth of the European drone services sector.
- Enabler for an estimated 25% reduction in delivery times for logistics and emergency services.

EI2 – Strengthening Digital Governance and Societal Acceptance

The municipal digital governance platform and evidence-based societal acceptance framework will foster democratic oversight, transparent data use, and citizen trust, ensuring that drone operations directly support EU values of security, privacy, and sustainability. This framework is projected to increase public acceptance of drone services by 40%, while enabling municipalities to reduce governance costs by 10–15% through streamlined digital oversight. Alignment with the EU Digital Decade goals ensures that drone services contribute to sustainable mobility, achieving estimated CO_2 savings of up to 25% per delivery compared to road transport.

KPIs & Economic Benefits:

- 50% increase in citizen trust/acceptance as measured in surveys across pilot cities.
- Reduction of urban logistics-related emissions by 25% CO_2 annually at EU scale.
- 15% cost savings in municipal oversight through automated monitoring and governance.
- Contribution to EU Green Deal targets by enabling climate-neutral urban logistics.

EI3 – Driving Economic Growth and Technological Sovereignty

Through validated B2B business models in the exploitation section, and scalable deployment pathways, dronnect will catalyse investment in cutting-edge drone technologies, strengthening Europe's technological sovereignty while unlocking new value chains. At full deployment, this is expected to generate better profits margin in new market revenues within the next decade, with a return on investment (ROI) of 3–5x for early adopters. By reducing delivery times by up to 50%, enabling zero-emission urban logistics, and creating 10,000+ skilled jobs in drone services, manufacturing, and digital infrastructure, the project will directly support Europe's transition to a climate-neutral, digitally empowered economy.

KPIs & Economic Benefits:

- 40% better growth projected market revenues within 10 years.
- 3–5x ROI for stakeholders deploying validated business models.
- 50% reduction in last-mile delivery times for goods and services.
- Creation of 10,000+ new green and digital jobs across EU cities.

2.3. SCALE AND SIGNIFICANCE

The following table addresses the scale and significance of the outcome

Dimension	Description	Beneficiaries
Overall objective	Bring together stakeholders of the drone logistics ecosystem (industry operators, logistics providers, investors, regulators, municipal authorities) and enable aviation authorities/policy makers to build regulatory certainty for fast-paced B2B deployment.	Industry, regulators, policy makers, municipal authorities
Societal engagement	Citizens' concerns addressed via surveys, immersive VR study experiments, public demonstrations, and round-table discussions, ensuring acceptance is systematically captured and translated into mitigation strategies.	Citizens, municipal authorities, regulators
Dual approach	Builds trust with the public while creating strong exploitation opportunities for industry and investors.	Citizens, industry, investors
Targeted engagement	Coordinated dissemination & communication actions to engage: • Policy makers (<50 across Europe) • Industry stakeholders (<200) • Society/general public (<2000) • Research entities & technology providers (knowledge transfer).	Policy makers, industry, citizens, research & technology providers

Dimension	Description	Beneficiaries
Strategic alignment	Contributes directly to EU Drone Strategy 2.0, European Green Deal, and Cities Mission objectives for climate-neutral, smart, and sustainable cities.	EU institutions, regulators, municipal authorities
Sustainability contribution	Supports sustainable mobility and efficient logistics, reducing emissions (Scopes 1 & 2) via drone logistics as a zero-emission alternative.	Citizens, municipalities, logistics providers
Key results	Validated frameworks for: • Resilient communication infrastructure • Seamless U-space integration • Municipal governance platforms • Societal acceptance strategies • Scalable business models.	Industry, regulators, municipal authorities, technology providers
Policy integration	Results integrated into City Climate Contracts (CCCs) and Sustainable Urban Mobility Plans (SUMPs), with tools and roadmaps for regulators and cities.	Municipal authorities, regulators, policy makers
Broader impact	Knowledge transfer through surveys, public demos, immersive experiments, and stakeholder round-tables, reaching thousands of citizens, policy makers, and market actors.	Citizens, policy makers, industry stakeholders
Long-term contribution	Supports large-scale uptake of drone logistics and long-term EU climate-neutral and digital transition goals.	EU institutions, citizens, industry, municipal authorities

2.4. POTENTIAL BARRIER

Political and Legal: Strong EU-level policy support through Drone Strategy 2.0, Horizon Europe Cluster 5 – Destination 6, and Cities Mission creates favourable conditions for urban drone integration. National aviation authorities (e.g., under EASA AMC-GM) increasingly harmonize certification, licensing, and U-Space rules, but variations in local governance and acceptance can delay deployment. Municipal and local authorities play a key role in granting permissions, utilizing and feedback on the digital tools, decision making, larger scope - integrating drone logistics into Sustainable Urban Mobility Plans (SUMPs). Deployment depends on evolving **EU drone regulations** (EASA), standardization of communication protocols, and liability frameworks. Data protection (GDPR) and cybersecurity regulations are critical for public trust in digital governance platforms. Given the expertise of the consortium the actions to mitigate these issues is already set in motion with relevant collaboration and exchange (WP 5-13). **Economic:** Economic factors are the significant in drone market, The drone services market is projected to reach **€14.5 billion by 2030**, with logistics as one of the strongest growth segments. Cost savings from drone logistics (fuel reduction, faster delivery) increase competitiveness of urban supply chains. Early market adoption requires significant investment in communication infrastructure, U-Space platforms, and operational frameworks; risk reduction is critical for private investors. As the consortium is aware and business, investment viability in tasks of WP5, WP14 will enable concrete evaluation before the end of project to provide suitable cost-effective business scaling (WP 15) and implementation strategy **Social:** Public acceptance is both a challenge and an opportunity: surveys and immersive VR experiments show citizens' concerns about **safety, privacy, and noise**, but also willingness to adopt drones for societal benefits (e.g., emergency response, medical logistics). **Trust-building measures** such as transparency tools, participatory governance, and real-world demonstrations will influence uptake (WP 9,10,12). Workforce impacts include creation of **new green jobs** in drone operations, maintenance, and urban mobility planning. The SSH expertise through WP 9, 10 will play a key role in the project to allow for greater consumer/user awareness and empowerment. **Technological:** Reliable multi-link communication infrastructure (5G/LTE, RF, satellite) is a prerequisite for safe BVLOS operations. Integration with U-Space/UTM systems requires interoperable, safe handover and standardized interfaces. Emerging capabilities in AI, automation, and digital twins enhance fleet management, predictive safety, and urban integration. VR-based public acceptance testing provides innovative tools to pre-assess societal impact before large-scale rollout. **Environmental:** *dronnect* directly supports the EU climate-neutral goals as a part of EU drone Strategy 2.0 target by reducing urban transport emissions through zero-emission drone logistics. Reduced congestion and optimized last-mile logistics improve air quality in dense urban areas. **Lifecycle considerations (e.g., battery production and disposal)** need mitigation strategies to ensure overall sustainability. The drone operator expert in the consortium considers and suggest potential measures through the WP 8 and tasks in technological pillar.

2.5. Measures to maximize impact – Dissemination, exploitation and communication #@COM-DIS-VIS-CDV@#

Effective communication, dissemination, and exploitation — supported by robust IPR and data management — are critical to achieving **dronnect's objectives** and ensuring its long-term impact. These activities will be strategically aligned across all work packages to form a coherent plan of action that not only promotes the project's results but also drives their uptake, scalability, and sustainability well beyond the project's lifetime. At the core of this approach lies the **Dronnect CDE Platform**, a modular framework composed of the **Sentiment Analysis Monitor (Communication)**, the **Research Impact Monitor (Dissemination)**, and the **Exploitation Radar (Exploitation)**. These tools will provide real-time insights into public perception, track the scientific and policy impact of project outputs, and systematically monitor the maturity and market readiness of Key Exploitable Results (KERs). By embedding this data-driven platform into the project's strategy, Dronnect ensures that communication, dissemination, and exploitation activities remain adaptive, evidence-based, and directly aligned with stakeholder needs and market dynamics.

Communication and Dissemination

The **dronnect project** will deliver a targeted and engaging dissemination strategy to ensure that its knowledge, tools and results are visible, accessible and relevant to the stakeholders who can apply them. This strategy will foster an ongoing dialogue with drone operators, regulators, cities, and citizens, helping technical and societal innovations translate into **safe, trusted and sustainable drone logistics services**. The dedicated Communication tasks will establish a clear and recognizable brand identity that reflects Dronnect's mission of *"Connecting Skies, Building Trust, Powering Growth."* This identity will underpin all communications, ensuring consistency across digital and print channels. Messages will be tailored to the needs of diverse audiences — operators, logistics providers, municipal authorities, regulators, investors, citizens and the research community — while

remaining fully aligned with the project's values of **safety, transparency and societal trust**. To achieve this, Dronnect will:
A) Develop a comprehensive **Communication and Dissemination Plan**, setting objectives, defining audiences, crafting key messages, and establishing performance indicators.
B) Build and maintain a strong **visual identity** and consistent messaging across all media.
C) Manage engaging **digital channels** — Project website, Newsletters, Social Media profiles (LinkedIn, , YouTube, X/Twitter) — guided by the **Sentiment Analysis Monitor**, which will continuously capture reach, tone, and engagement.
D) Publish and present results in **high-impact journals** (e.g., *Drones, Aerospace, Journal of Air Transport Management*) and at **leading conferences** (e.g., *Amsterdam Drone Week, SESAR Innovation Days, EU Drone Days*). Dissemination impact will be tracked through the **Research Impact Monitor**.
E) Ensure participation in **flagship dissemination events,** which combine technical showcases, citizen engagement and policy dialogues.
F) Engage with **expert networks and working groups** (e.g., SESAR 3 JU, EASA IAM Hub, EUROCAE) to strengthen dronnect's voice in the European drone and innovative air mobility ecosystem.

Dissemination efforts will remain dynamic, informed by analytics and feedback to ensure ongoing relevance and impact. The combination of public-facing communication, targeted stakeholder outreach, and rigorous scientific dissemination will maximize dronnect's visibility and uptake. At the heart of this approach will be **personalized, multi-channel communication**. dronnect will strengthen connections with its audiences by delivering tailored messages across multiple touchpoints. This will be supported by the **Sentiment Analysis Monitor**, which uses AI-powered analytics and natural language processing to monitor public perceptions and stakeholder priorities in real time, enabling continuous refinement of messaging and preparing the ground for stakeholder engagement activities. **Stakeholders' engagement :** dronnect will pursue a **dynamic, evidence-driven engagement strategy**, starting with a clear understanding of the priorities, behaviors, and concerns of its audiences. Building on the consortium's expertise in **U-space, governance, and societal acceptance**, the project will apply advanced communication practices and AI-powered analysis to establish an initial baseline of public sentiment around its thematic areas.The **Sentiment Analysis Monitor** is a dedicated tool for capturing and analyzing **public sentiment across social media platforms**. Its core objective is to extract actionable insights from digital conversations to inform and continuously refine the project's dissemination and communication strategy as well as track user acceptance and adoption of technologies. This monitor acts as a **real-time barometer** of how a project is perceived online, focusing on social media and web platforms such as **X (formerly Twitter), LinkedIn, Facebook, YouTube, Reddit**, and relevant forums and blogs. By analyzing both **quantitative trends** (e.g., volume, reach, engagement) and **qualitative signals** (e.g., tone, opinion clusters, emotional responses), it helps the consortium understand how the project's priorities are being received, debated, or supported. Innovation Dis.Co deploys advanced **AI-based natural language processing (NLP)** and **text mining** techniques to:

- Track evolving **public opinion**, attitudes, and concerns related to the project's key themes.
- Identify **influential voices** and stakeholder groups engaging with the project.
- Detect emerging **narratives, opportunities, and reputational risks**.

The Sentiment Analysis Monitor is built around three core pillars:

- **Social Influence**: Mapping the spread of project-related content and its amplification through digital networks.
- **Text Mining**: Extracting relevant data from high-volume social media posts, comments, and shares.
- **Sentiment Analysis**: Interpreting tone, emotion, and polarity (positive, neutral, negative) to assess public perception over time.

Ultimately, this tool enables projects to make **data-driven communication decisions**, engage more effectively with online audiences, and respond proactively to shifts in public discourse. dronnect places stakeholder engagement at the heart of its mission to advance drone-enabled logistics. From the outset, the consortium has focused on understanding how different audiences, ranging from policymakers and regulators to logistics providers, industry associations, and the general public, perceive both the opportunities and challenges of drone adoption. To achieve this, Innovation Dis.Co leveraged state-of-the-art communication methods combined with AI-driven analytics through its

Sentiment Analysis Monitor platform, establishing an initial baseline of public perception. Between June and September 2025, a comprehensive monitoring campaign was carried out to capture real-world discussions and stakeholder attitudes.The campaign registered **8.430 relevant mentions** across social and traditional media, with a combined **reach of 5.4 million people** and over **160,000 interactions**. The analysis examined how citizens, industry professionals and regulatory bodies currently discuss and perceive issues related to drone logistics, U-Space integration, BVLOS operations, drone acceptance, municipal drone governance and the EU Drone Strategy 2.0. Particular attention was given to themes of trust, safety and societal readiness to adopt drone-enabled logistics. Results indicate that the monitored discourse reflects a strong **positive predisposition (82%)** toward innovation and expert engagement, while **15% of mentions expressed concerns**, mainly around regulation, safety and social acceptance while only **3% remained neutral**. These insights confirm that the sector is viewed as promising, yet still requires active dialogue and reassurance on governance and societal impact. The analysis also revealed trending themes and hashtags such as **#dronedelivery, #logistics, #innovation, #bvlos**, as well as key influencers and platforms shaping the debate (LinkedIn, YouTube, Reddit, and X/Twitter). This mapping provides Dronnect with a solid foundation to design targeted engagement activities and tailor its communication to each stakeholder group. Throughout the project, for each tracked keyword, sentiment polarity will be interpreted as a proxy for ecosystem readiness, barriers and opportunities. Positive sentiment reflects adoption drivers, negative sentiment highlights risks to be mitigated, and neutral sentiment provides baseline monitoring of factual developments. This helps the consortium and involved stakeholders see **what positive, negative, and neutral sentiment trends actually mean** in the context of **dronnect's impact.** In addition to providing a baseline of attitudes, the monitoring **will continuously inform the project's strategy by highlighting emerging themes, influential voices, and priority channels.** This enables dronnect to refine its messaging in real time, anticipate risks and measure progress against dissemination and communication objectives. Building on this evidence, dronnect will implement a **layered engagement approach** that combines monitoring, outreach and co-creation:

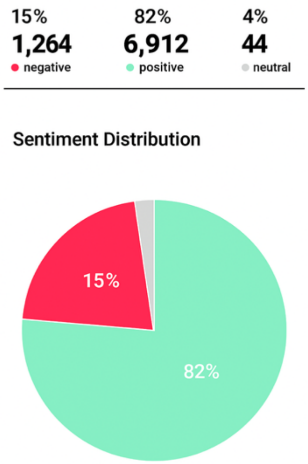


Figure on Public Sentiment Breakdown

- **Strengthening policy dialogue** with regulators and aviation authorities to inform evolving BVLOS frameworks and drone safety standards.
- **Creating synergies with industry stakeholders**, including logistics operators, drone providers and technology companies, to accelerate adoption and integration.
- **Organising interactive events** such as live demonstrations, workshops and policy roundtables to foster two-way knowledge exchange.
- **Leveraging continuous monitoring** to refine messaging in real time, ensuring communication remains relevant to public perceptions and emerging debates.
- **Nurturing trust** by addressing societal concerns and showcasing the added value of drone logistics, emergency response, and sustainable transport.

The overall objective is to **expand the size, reach, and vibrancy** of the dronnect ecosystem. This will strengthen both the scientific and societal impact of the project and ensure its results continue to deliver value during and beyond its lifetime. **Research Impact monitoring:** Monitoring scientific impact in a Horizon Europe project is essential to not only meet EU dissemination obligations but to also track and demonstrate the value of research outputs. It helps track visibility through publications, citations and online engagement, informs stakeholders, supports communication and exploitation activities, and improves outreach strategies. Ultimately, it ensures the project's results are used, recognized and are impactful beyond the consortium.

To that end, the **Research Impact Monitor** supports:

- **Researchers impact tracking:** Tracking of the project partners' dissemination & scientific reach
- **Citation analysis:** Tracking and analyzing total citations, downloads and readership of project publications (e.g., on platforms such as Mendeley etc.).

- **Online engagement monitoring:** Monitoring online engagement by capturing research mentions across social media, news websites, policy documents, and blogs.
- **Visualization**: Visualizing the dissemination footprint of project research in real-time.

Further (optional) tools include:

- ***Co-Citation Analysis****: Identifying relationships between dronnect research and other highly cited works, providing insight into research networks and influence.*
- ***Bibliographic Coupling****: Analyzing connections between dronnect publications and other studies citing the same references, highlighting thematic clusters and emerging trends.*

Planned dissemination and communication measures & activities: dronnect will aim to showcase its solutions and activites at leading international conferences and industry events where consortium partners already have a strong presence and can attract key stakeholders in aviation, logistics and urban mobility. In parallel, the partners will leverage their professional networks to directly engage with drone operators, municipalities, logistics providers and prospective investors. Indicatively:

Target Journals	Key Conferences & Events
Drones **– Open access, high-impact in UAV technologies, operations, and applications.**	Amsterdam Drone Week – Europe's flagship event on U-space and drone integration, including a dedicated Dronnect session.
Journal of Air Transport Management **– Focus on aviation management, safety, and policy.**	SESAR Innovation Days – Premier EU forum for air traffic management research and innovation.
Aerospace **– Covers aerospace engineering, communication systems, and UAV applications.**	EASA High Level Conference on Drones – High-level policy dialogue on drone regulation and public acceptance.
Transportation Research Part C **– Focus on intelligent transport systems and digital integration.**	World ATM Congress – Global gathering on air traffic management, relevant for U-space and BVLOS integration.
Safety Science **– Multidisciplinary journal bridging safety, human factors, and risk management.**	EUROCAE Symposium – Standardisation platform on aviation systems and interoperability.
Journal of Intelligent & Robotic Systems **– Robotics, autonomy, and AI in UAVs.**	DroneX Expo – Among other topics, focuses into integrating drone logistics and mobility solutions in cities.
Remote Sensing **– Applied to UAV sensing, monitoring, and airspace management.**	Smart City Expo World Congress – Forum on multimodal mobility and municipal integration of drones.
	Amsterdam Drone Week – Europe's flagship event on U-space and drone integration, including a dedicated Dronnect session.

To fully support dissemination and maximize the project's impact — particularly given its ambition to demonstrate safe, scalable, and publicly trusted **drone logistics operations** — Dronnect will establish a well-structured and far-reaching set of communication activities and tools. These actions will amplify outreach to targeted scientific communities, logistics operators, aviation authorities, policymakers, municipalities, and citizens. To achieve this, dronnect will implement a coordinated series of actions to present project results in a **tangible, accessible, and compelling** manner for diverse audiences, while running **targeted campaigns** tailored to their needs. Specifically, Dronnect will:

1. Deploy innovative communication tools and channels as outlined in the **Communication and Dissemination Plan**, ensuring clear, consistent messaging across all outputs.
2. Leverage each partner's communication infrastructure — including institutional websites, newsletters, media contacts, and social media — amplified by **Sentiment Analysis Monitor** insights to refine messaging in real time.
3. Utilize European Commission media and dissemination services, ensuring visibility on **CORDIS, the Horizon Results Platform, and related EU outreach channels**.

4. Foster an active stakeholder community, connecting drone operators, logistics companies, regulators, cities, and citizens through both **digital and face-to-face engagement**.

The choice of tools and formats — including **co-creation techniques, immersive VR engagement, visual storytelling, multimedia content and dynamic social media strategies** — will be reviewed at project outset and refined in the first dedicated Dissemination & Communication deliverable. Continuous monitoring through the **Sentiment Analysis Monitor** will allow the consortium to identify emerging narratives, adjust outreach and ensure measurable impact.

A set of **core actions** will anchor this strategy:

- **Project website** – A continuously updated, central hub hosted and managed by the Coordinator, providing open access to non-confidential results, news, and event information. The website will act as the main entry point for stakeholders to connect with Dronnect, ensuring visibility of project activities and outcomes.
- **Social media monitoring and promotion** – Active project profiles will be maintained on selected platforms (LinkedIn, X/Twitter, YouTube), supported by targeted campaigns to boost awareness and engagement. A "live" dashboard powered by the **Sentiment Analysis Monitor** will integrate ongoing e-listening and analytics, enabling real-time tracking of public and professional perceptions on Dronnect's themes such as drone logistics, U-space, and societal acceptance.

Beyond digital outreach and monitoring, Dronnect will deploy a range of tailored communication tools and activities to engage different audiences effectively:

- **Press releases** – Issued at strategic milestones to highlight major achievements, breakthroughs, or events with high societal or market relevance. Content will be adapted for general and specialist media, distributed via consortium channels and European Commission dissemination services.
- **Promotional materials** – Brochures, flyers, posters, infographics, technical briefs, videos, webinars, and stakeholder-focused presentations will be created to meet the needs of specific audiences. Where appropriate, translation and localization will be applied to support wider uptake. Videos will be distributed through the project's YouTube channel, partner websites, and sector-specific platforms.
- **Project newsletter** – Starting in Month 6, a biannual e-newsletter will provide project updates, highlight results, and announce upcoming events and engagement opportunities. It will be disseminated via the website, mailing lists and social media channels.
- **Targeted presentations** – Customized presentations will be prepared for scientific, policy, community and industry audiences, ensuring relevance, clarity, and immediate applicability.
- **Scientific publications** – Peer-reviewed articles will disseminate Dronnect's technical, societal, and governance results. In line with Horizon Europe's open science requirements, the consortium will aim for green open access as a minimum, pursuing gold open access whenever possible. Uptake and reach will be tracked through the **Research Impact Monitor**.
- **Standardisation activities** – Where relevant, Dronnect partners will work with bodies such as **EASA, SESAR 3 JU, and EUROCAE** to support the integration of project results into emerging or existing standards, ensuring long-term adoption and impact.

Key performance indicators for dissemination and communication: Defining **key performance indicators (KPIs)** to track dissemination and communication progress is essential to closely monitor activities and measure their effectiveness. For Dronnect, KPIs will be continuously monitored and managed by the WP leader, with corrective measures applied whenever performance falls short of objectives. This ensures that communication, dissemination, and exploitation activities remain fully aligned with the project's overall impact pathway. The **Sentiment Analysis Monitor** will complement traditional analytics by assessing tone, reach, and stakeholder sentiment across social media, while the **Research Impact Monitor** will track scientific visibility through citations, downloads, and online engagement. Together with standard web and media analytics, these modules will provide a robust evidence base for real-time adjustments.

Measure	Indicators	Target	Time-line	Means of Verification
Dronnect website	Number of unique visitors	≥ 10,000 total	By M36	Analytics reports

Measure	Indicators	Target	Time-line	Means of Verification
	Number of page views	≥ 15,000 total	By M36	Analytics reports
Social networks	Number of followers across LinkedIn, X/Twitter, YouTube	≥ 2,000 total	By M36	Official platform analytics + Sentiment Analysis Monitor
	Number of reaches	≥ 100,000 total	By M36	Platform analytics + Sentiment Analysis Monitor
	Average engagement rate per post	≥ 5% per post	Continuous	Platform analytics + Sentiment Analysis Monitor
Dronnect workshops	Number of participants in project-organized workshops	≥ 200 participants (total)	≥ 3 events by M36	Attendance records, media coverage
Videos	Number of videos published	6 total	By M36	YouTube/website analytics
	Average number of views per video	≥ 1,000 views/video	Continuous	Platform analytics
Scientific publications	Number of peer-reviewed papers/articles	≥ 10	By M36	DOIs, repository entries + Research Impact Monitor
Dronnect brochures	Number of brochures distributed (digital + print)	≥ 1,000	By M36	Event proceedings, photos
Posters	Number of posters produced for events	≥ 4	By M36	Original poster files, event records
Project materials downloads	Downloads of resources (brochures, reports, toolkits)	≥ 500	By M36	Website analytics
Newsletters	Number of distributions	≥ 1,200	By M36	Mailing list statistics
Media outreach	Number of press releases issued and published	≥ 4	By M36	Media clippings, press records
External events participation	Number of external events (conferences, policy meetings, exhibitions) where Dronnect partners are present	≥ 12	By M36	Event records, programmes

Exploitation Strategy

The exploitation of **dronnect's Key Exploitable Results (KERs)** will be a central activity to ensure that the project's innovations — from resilient multi-link communication stacks to U-space operational toolkits, municipal governance platforms, societal acceptance frameworks, and B2B deployment models — generate lasting benefits beyond the project's duration. The exploitation strategy will be fully aligned with Dronnect's dissemination and communication plan, ensuring that all impact-maximization activities are integrated and mutually reinforcing.

The **Exploitation Radar** (as part of the dronnect CDE Platform) will support this process by providing a structured, visual methodology to track the maturity, adoption readiness and impact pathways. It provides a visual and dynamic framework to track progress, assess value, and prioritize innovation outcomes across technical, commercial, and societal dimensions.

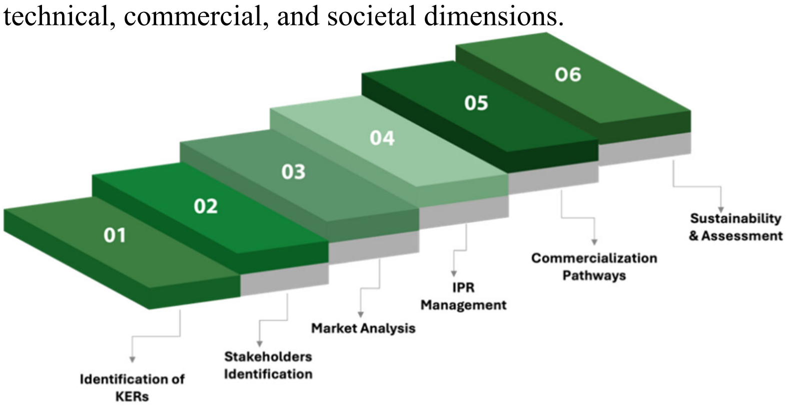


Figure on Exploitation Strategy main building blocks

<u>Governance and Coordination</u>

An **Exploitation Team** will be established at project start, composed of exploitation managers nominated by each partner. These managers will serve as the focal points within their organizations for exploitation matters, ensuring active participation in planning, monitoring, and executing exploitation activities. The Exploitation Team will:

- Coordinate exploitation activities across all work packages.
- Integrate exploitation considerations into technical, governance, and stakeholder engagement work from the outset.
- Oversee the alignment of exploitation plans with the project's IPR and data management strategy.
- Monitor and review exploitation opportunities using the **Exploitation Radar**, adjusting the strategy in response to market shifts, regulatory developments, and stakeholder needs.

Partner-Level and Consortium-Level Planning

Each partner will prepare and maintain an **Individual Exploitation Plan**, describing how they will use, adapt, or sustain specific KERs at the local, regional, or international level. These plans will also highlight opportunities for **co-exploitation** with other partners. The **Consortium-Level Exploitation Plan** will consolidate all partner-level inputs into a unified roadmap, reflecting the project's collective ambition to scale results. This plan will be updated regularly, incorporating feedback from pilots, stakeholder engagement, dissemination activities, and regulatory/market analyses.

Exploitation objectives

The exploitation strategy of Dronnect will follow three main stages of expansion, each with objectives designed to maximize impact and sustainability: **Short-term objectives (project duration)**

This stage begins with the launch of project activities and continues until the project's conclusion. The main aims are to: - Validate and demonstrate the feasibility, quality, and relevance of all KERs through pilot demonstrations in Hamburg and Rotterdam. -Refine communication systems, governance platforms, and societal frameworks based on stakeholder feedback and user testing. -Establish initial uptake pathways by integrating KERs into pilot logistics and emergency response scenarios. -Engage early adopters, including drone operators, municipal authorities, regulators, and logistics providers, to create momentum for post-project deployment. -Strengthen exploitation readiness by addressing IPR, licensing, and sustainability considerations during development. **Medium-term objectives (0–3 years after project end)**

This stage builds on the foundations established in the short term and focuses on expanding the reach and maturity of project outcomes: - Fully deploy mature KERs in participating cities and expand adoption to additional European contexts. - Fine-tune offerings (e.g., B2B toolkit, governance platform) based on early market experiences, performance monitoring, and evolving regulatory needs. - Enhance interoperability of dronnect's solutions with U-space and UTM platforms, as well as logistics IT systems, to support scale-up. - Extend partnerships and funding channels, including public-private partnerships, new grants, or service agreements, to sustain and expand implementation. - Promote uptake through training and capacity-building, ensuring that city staff, drone operators, and logistics providers become multipliers of Dronnect innovations.

1. **Long-term objectives (after project end)**

This stage aims to achieve sustained, large-scale adoption of project results: - Integrate dronnect solutions into national and European regulatory frameworks, ensuring permanent inclusion in **Sustainable Urban Mobility Plans (SUMPs)** and **City Climate Contracts (CCCs)**. - Achieve operational and financial sustainability for key KERs (e.g., the governance platform and B2B models) through established business models, licensing, or hybrid funding mechanisms. - Expand the geographical footprint to additional EU cities and cross-border corridors, adapting to local governance and mobility ecosystems. - Contribute to EU leadership in drone logistics by maintaining Dronnect as a reference model for resilient communications, U-space integration, and societal trust frameworks. - Continue innovation cycles by leveraging datasets, governance models, and stakeholder networks generated by Dronnect to feed into further research, development, and standardisation initiatives.

Planned exploitation measures & activities

dronnect's exploitation strategy will be supported by a clear set of measures, ensuring that Key Exploitable Results (KERs) are validated, protected, and positioned for long-term uptake. The **Exploitation Radar** of the CDE Platform will provide structured, real-time monitoring of each KER's maturity, market readiness, and exploitation pathway.

KER Identification and Refinement – Continuous mapping of exploitable assets throughout the project lifecycle, supported by the **Exploitation Radar**, which will track progress across dimensions such as TRL, market maturity, and stakeholder adoption. **Market and Context Analysis** – Initial and final analyses of market, regulatory, societal, and financial factors influencing adoption. This will include segmentation of target users (cities, operators, regulators, logistics providers) and the identification of emerging trends, informed by data from the **Sentiment Analysis Monitor**. **IPR Management** – Development of a consortium-wide IPR strategy, aligned with the Consortium Agreement, ensuring clarity on ownership, licensing, and revenue-sharing for joint and individual exploitation. The strategy will also guide which KERs require legal protection (e.g., patents, copyrights) versus those best disseminated as open knowledge. **Risk Management** – Establishment of a proactive exploitation risk strategy addressing technical, financial, market, and policy-related risks. This will include continuous monitoring of societal acceptance through the **Sentiment Analysis Monitor**, and alignment with data protection and certification requirements. **Exploitation Models** – Exploration and definition of viable business and sustainability models, including licensing agreements, software-as-a-service (SaaS) models for municipalities, and public–private partnerships for scaling drone logistics services. **Stakeholder Relationships** – Early and sustained engagement with high-influence stakeholders such as national aviation authorities, EASA, SESAR 3 JU, EUROCAE, city administrations, drone operators, and logistics providers. These relationships will accelerate adoption, validate governance tools, and secure Letters of Intent (LoIs) for post-project exploitation. **Validation Campaigns** – Testing and validation of exploitation pathways during pilot implementations in Hamburg and Rotterdam, ensuring readiness for scale-up and real-world deployment. This includes citizen acceptance testing via VR demonstrations and municipal adoption pathways through governance workshops.

Sustainability Commitment

By combining **partner-level exploitation planning** with a coordinated consortium-wide strategy, Dronnect will ensure that each KER is positioned for sustained impact — whether through commercialization, licensing, municipal integration, open-access dissemination, or incorporation into European drone policy and U-space frameworks. The **Exploitation Radar** will be continuously updated to provide evidence-based tracking of KER uptake and sustainability pathways. This approach will maximize return on investment for the EU while guaranteeing that dronnect's innovations continue to benefit cities, regulators, drone operators, and citizens well beyond the project's lifetime. The **dronnect consortium** recognizes that effective management of knowledge and Intellectual Property Rights (IPR) is critical for smooth collaboration among partners and for maximizing the long-term value, exploitation, and sustainability of project results. By ensuring transparent knowledge management and safeguarding partners' interests, the consortium will prevent information bottlenecks, respect confidentiality, and increase the likelihood of high visibility, market uptake, and regulatory adoption of dronnect's outputs. IPR and knowledge management will be embedded in the framework of the **Consortium Agreement (CA)**, which will be fully aligned with European Commission policies and Horizon Europe best practices. These provisions will be operationalized through an **IPR Management Plan**, developed during the project's early phase, ensuring clarity and shared understanding across all partners. The CA will explicitly define how partners may access pre-existing (Background) and newly generated (Foreground) knowledge from other beneficiaries, both during the project and for exploitation beyond its lifetime.

The CA will address in detail: **Confidentiality** – rules for sharing and protecting sensitive information, in compliance with applicable laws and EU regulations. **Ownership of results** – clear definition of which partner(s) own specific outputs and under what conditions. **Legal protection of results** – including patents, copyrights, database rights, or other forms of intellectual property protection. **Access rights** – terms for accessing both Background and Foreground knowledge for research, development, or exploitation. **Obligation for use** – partners' responsibilities to ensure results are used in line with the Grant Agreement. **Dissemination of knowledge** – ensuring that all communication and dissemination activities comply with IPR rules and the Horizon Europe Model Grant Agreement.

The consortium will also follow **Open Science principles** to ensure that dronnect's scientific and technical outputs — including peer-reviewed papers, algorithms, datasets, software tools, and governance models — are made accessible to the research and innovation community. For scientific publications, **green open access** will be prioritized, with final peer-reviewed versions deposited via the project website and trusted repositories such as **Zenodo, Open Research Europe, and OpenAIRE**. Where feasible and within budget constraints, **gold open access** will also be pursued. The **Research Impact Monitor** will be used to track visibility, citations, and dissemination footprint, ensuring that published results achieve the widest possible reach. In addition, the consortium will establish **clear procedures for the legal and ethical sharing of datasets**, ensuring compliance with **GDPR** and other data protection regulations. This will safeguard survey results, VR acceptance studies, and network performance data collected during pilots. The **Exploitation Radar** will complement these measures by mapping which Key Exploitable Results (KERs) require formal protection (patents, licensing) versus those that should remain openly accessible to maximize adoption. Through this integrated approach, dronnect ensures that its outputs — such as the multi-link communication stack, U-space procedures, municipal governance platform, acceptance framework, and B2B models — can be sustainably accessed, protected, and built upon by future projects, regulators, industry stakeholders, and policymakers. #§COM-DIS-VIS-CDV§#

2.6. Summary

Specific Needs	Expected Results	D & E & C Measures
What are the specific needs that triggered this project? Communication Infrastructure Gap: Absence of robust, resilient multi-link communication systems for BVLOS drone operations in urban environments Regulatory Integration Challenges: Lack of seamless integration between U-space and non-U-space airspace operations Societal Acceptance Barriers: Limited understanding of public concerns regarding safety, noise, privacy, and operational transparency in drone logistics Municipal Governance Gap: Missing digital platforms for municipal authorities to actively participate in drone traffic management Economic Uncertainty: Unclear return on investment (ROI) and viable B2B business models for drone logistics deployment Operational Readiness: Insufficient validated frameworks for take-off/landing sites, emergency procedures, and contingency management in VLL corridors	**What do you expect to generate by the end of the project?** Resilient Multi-Link Communication Infrastructure with ICAO Required Communication Performance practical (low margin and higher margin) > 95% availability and API response times Validated Operational Framework for seamless U-space integration with >70% emergency procedure success Municipal Digital Governance Platform deployed in 3+ cities with >70% engagement rates Evidence-Based Public Acceptance Framework from EU-wide survey respondents and VR validation Economically Viable B2B Drone Logistics Models with proven ROI across 5+ use cases Technical Roadmaps for standardization of communication infrastructure and traffic management systems Policy Recommendations and mitigation strategies for responsible drone deployment	**What dissemination, exploitation and communication measures will you apply to the results?** Digital Channels: Project website, social media (YouTube, LinkedIn), periodic newsletters Scientific Publications: Open access publications, conference presentations, technical reports Stakeholder Engagement: Round-table discussions, workshops, webinars, pilot demonstrations Training Materials: Best practice handbooks, competence-building materials, replication strategies Public Events: "Dronnect Demo Day" with live demonstrations and stakeholder feedback sessions Media Outreach: video storytelling, press releases, digital content creation Business Development: B2B model validation, stakeholder surveys, market analysis

Target Groups	Outcomes	Impacts
Who will use or further up-take the results of the project? Who will benefit from the results of the project?	**What change do you expect to see after successful dissemination and exploitation of project results to the target group(s)?**	**What are the expected wider scientific, economic and societal effects of the project contributing to the expected impacts outlined in the respective destination in the work programme?**
Citizens: General public, urban communities, peri-urban residents, inter-modal transport users **Logistics Operators:** Warehouses, hospitals, laboratories, supply chains, last-mile service providers **Drone Operators:** Service providers, fleet management operators, technology enablers, manufacturers **ATM Providers:** Air Traffic Management, UTM providers, USSP, CISP **Government & Authorities**: Municipalities, regulatory agencies (EASA, SESAR JU), standardization bodies **External Investors**: Venture capital, corporate investors, public funding agencies, insurance entities	Enhanced Operational Reliability: Drone operators achieve uninterrupted BVLOS connectivity and standardized workflows Improved Municipal Governance: Authorities enable transparent decision-making through digital platforms Increased Public Acceptance: Citizens adopt drone logistics more readily through evidence-based engagement strategies Streamlined Regulatory Processes: Government authorities implement informed policy measures and achieve streamlined approvals Economic Viability: Logistics operators achieve 50% faster deliveries with proven ROI models Investment Confidence: External investors benefit from reduced societal risk and validated business models Market Readiness: Transition from pilot projects to commercial-scale operations across EU cities	EU Drone Strategy 2.0 Acceleration: Contributing to €14.5 billion drone services market by 2030 and 145,000 jobs creation Digital Democracy Enhancement: Supporting Digital Decade goals Climate Neutrality Contribution: supporting EU Green Deal objectives Technological Sovereignty: Strengthening Europe's leadership in drone technologies and reducing strategic dependencies Green Jobs Creation: New employment opportunities in climate-neutral drone operations and urban mobility planning Cross-Border Harmonization: Validated compliance procedures supporting regulatory approval across EU Member States (Drone Strategy) Smart city goals

#§IMP-ACT-IA§#

3. QUALITY AND EFFICIENCY OF THE IMPLEMENTATION #@QUA-LIT-QL@# #@WRK-PLA-WP@#

WORK PLAN AND RESOURCES

BRIEF PRESENTATION OF THE OVERALL STRUCTURE OF THE WORK PLAN

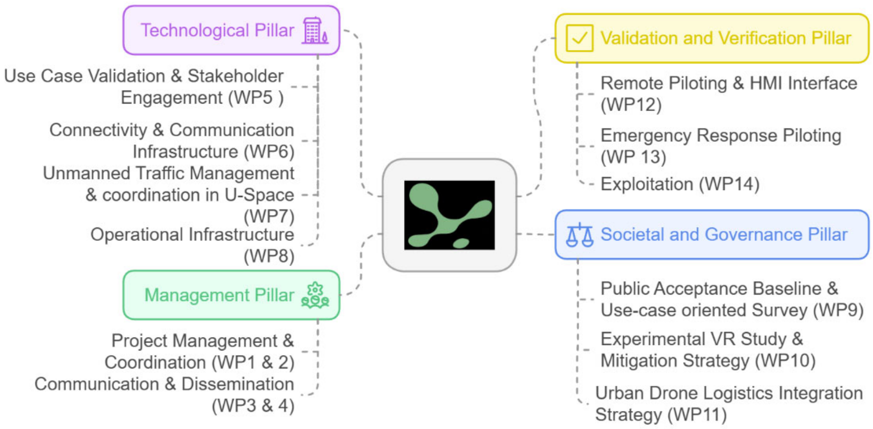


Figure3.1: Representation of dronnect pillars and associated WPs

The pillars and their associated Work Packages (WPs) – figure 3.1, which define the tasks and activities required to achieve the project objectives (as outlined in Section 1), are explained in detail in this section**. In total, 15 work packages are distributed across four pillars, providing a structured framework to ensure the successful execution of the project.** The following table presents the Gantt chart for *dronnect*, outlining the respective WPs and tasks distributed over a 3-year period (36 months). **The pilot demonstration, as described in Section 1, is scheduled for Year 3, Quarter 3 (Months 30–33)**

3.1.1. GRAPHICAL PRESENTATION OF THE COMPONENTS SHOWING HOW THEY INTER-RELATE

The project is structured around four main pillars, each consciously designed to address the overall objectives and provide answers to the research questions, ultimately leading to the planned outcomes. A total of 15 work packages (WPs) are distributed across these pillars, ensuring a coherent and goal-oriented approach. Collaboration across tasks is strategically planned to align with the project timeline and ensure timely delivery of outputs. Pillar 1: Management (WP1–WP5) – Oversees coordination, administration, quality assurance, and overall project governance. Pillar 2: Technological Development (WP6–WP8) – Focuses on the design, development, and integration of technological solutions. Pillar 3: Societal and Governance (WP9–WP11) – Addresses ethical, societal, and regulatory dimensions, ensuring alignment with broader societal needs and compliance frameworks. Pillar 4: Verification and Validation (WP12–WP15) – Ensures systematic testing, evaluation, and validation of outcomes against defined objectives.

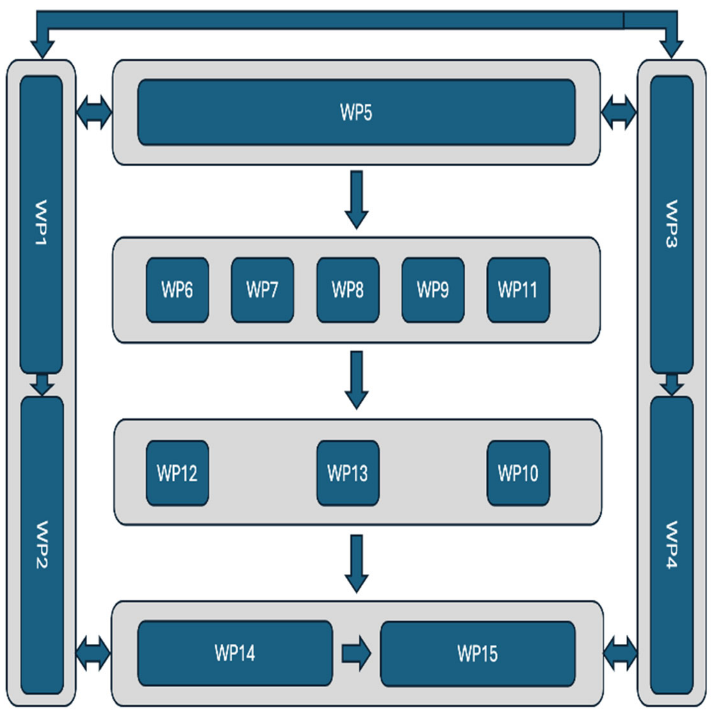


Figure3.2: Dependencies of WPs in pert chart

The accompanying chart illustrates the workflow and interdependencies across these pillars, demonstrating how each contributes to the overall objectives while remaining interconnected to ensure smooth progress toward the final outcomes.

TIMING OF THE DIFFERENT WORK PACKAGES AND THEIR COMPONENTS

Quality and efficiency of the implementation #@QUA-LIT-QL@# #@WRK-PLA-WP@#-2: Timing of the different work packages and their component

WP	Task	Title	Lead	Y1 Q1	Y1 Q2	Y1 Q3	Y1 Q4	Y2 Q1	Y2 Q2	Y2 Q3	Y2 Q4	Y3 Q1	Y3 Q2	Y3 Q3	Y3 Q4
		Milestones			M1, M2				M3		M4			M5	M6
WP 1		PROJECT MANAGEMENT & COORDINATION RP1	TUD	D1.1	D1.2				D1.3						
	T1.1	Scientific and technical coordination	TUD												
	T1.2	Administrative management	TUD												
	T1.3	Gender, ethics, data management	TUD												
WP2		PROJECT MANAGEMENT & COORDINATION RP2	TUD												
	T2.1	Scientific and technical coordination	TUD												
	T2.2	Administrative management	TUD												
	T2.3	Gender, ethics, data management	TUD												
WP3		COMMUNICATION & DISSEMINATION RP1	IVD	D3.1,3.2	D3.3				D3.4						
	T3.1	Project identity and online channels	IVD												
	T3.2	Dissemination and communication strategy and implementation	IVD												
	T3.3	Communication and dissemination tools	IVD												
WP4		COMMUNICATION & DISSEMINATION RP2	IVD												D4.1
	T4.1	Project identity and online channels	IVD												
	T4.2	Dissemination and communication strategy and implementation	IVD												
	T4.3	Communication and dissemination tools	IVD												
WP5		USE CASE VALIDATION & STAKEHOLDER ENGAGEMENT	ANYWI		D5.1										
	T5.1	Definition of the use cases and specification requirements	ANYWI												
	T5.2	Initial stakeholder engagement survey	IVD												
	T5.3	Pilot design and demonstration planning	ANYWI												
WP6		CONNECTIVITY & COMMUNICATION INFRASTRUCTURE	TUD						D6.1						
	T6.1	Design of link architecture (drone-several entities) and associated protocols	TUD												
	T6.2	Implementation and validation of the communication link testbed	TUD												
	T6.3	Implementation of network performance monitoring dashboard	ANYWI												
	T6.4	Evaluation of the testbed and resource management optimization	ANYWI												
WP7		UNMANNED TRAFFIC MANAGEMENT & COORDINATION IN U-SPACE	HHLAS			D7.1							D7.2		
	T7.1	Identify challenges of connecting U-space and non-U-space BVLOS logistics operations	HHLAS												
	T7.2	Develop, refine, validate workflows, solutions, and guidelines for seamlessly entering, exiting, and traversing U-spaces in BVLOS operations	HHLAS												
WP8		OPERATIONAL INFRASTRUCTURE	BYV			D8.1					D8.2				
	T8.1	Identify challenges and strategy for preparation of take off and landing sites	BYV												
	T8.2	Analyze impact of disruptions in U-space through DAR issuance by ATM	HHLAS												
	T8.3	DAR: Develop and test strategies, guidelines for simultaneous landing of multiple UAS	HHLAS												
WP9		PUBLIC ACCEPTANCE BASELINE & USE CASE ORIENTED SURVEY	HFC		D9.1				D9.2						
	T9.1	Desk research on societal acceptance factors for drone logistics use case	HFC												
	T9.2	Development of the use case-oriented survey	HFC												
	T9.3	Empirical analysis of public acceptance by use case	HFC												
WP10		EXPERIMENTAL VR STUDY & MITIGATION STRATEGY	HFC												
	T10.1	Adaption of drone acceptance model to logistics use cases	HFC												
	T10.2	VR-based study development and execution	HFC												
	T10.3	Analysis of VR-based study and preparation of mitigation strategies	HFC												
WP11		URBAN DRONE INTEGRATION STRATEGY	TCN											D11.2	
	T11.1	Identification of policy gaps and model development of drone integration model for urban and peri-urban eco-system	TCN												
	T11.2	Development and implementation of the application platform to involve municipal governance	TCN												
	T11.3	Analysis and strategy development for smooth integration in urban multi-modal transportation ecosystem	TCN												
WP12		REMOTE PILOTING AND HMI INTERFACE	HHLAS						D12.1					D12.2	
	T12.1	Drone logistics pilot demonstrator: preparation and final demonstration	HHLAS												
	T12.2	Feasibility study of HMI interface (one remote pilot to multiple drones)	HHLAS												
WP13		EMERGENCY RESPONSE PILOTING	BYV											D13.1	
	T13.1	Emergency response demonstrator: preparation and final demonstration	BYV												
	T13.2	Analysis and lessons learnt and strategies to systemize the operation	BYV												
WP14		EXPLOITATION RP 1	IVD						D14.1						
	T14.1	Exploitation plan and implementation	IVD												
	T14.2	Stakeholder study and B2B drone logistics business model	IVD												
WP15		EXPLOITATION RP 2	IVD								D15.1			D15.2	D15.3, 15.4, 15.5
	T15.1	Roadmap towards communication infrastructure and traffic management system standardization	ANYWI												
	T15.2	Analysis of suitable model and strategies and tools for fast paced deployment	IVD												
	T15.3	Public acceptance and social Impact and mitigation strategies for drone logistics usecase	HFC												
									RP1	RP2					

3.1.2. DETAILED WORK DESCRIPTIONS

Table Quality and efficiency of the implementation #@QUA-LIT-QL@# #@WRK-PLA-WP@#-1: List of work packages

WP No	Work Package Title	Lead No	Lead Partner	Person Months	Start Month	End Month
WP 1	Project Management & Coordination RP1	1	TUD	16.5	01	18
WP 2	Project Management & Coordination RP2	1	TUD	19.5	19	36
WP 3	Communication & Dissemination RP1	7	IVD	23.5	01	18
WP 4	Communication & Dissemination RP2	7	IVD	24	19	36
WP 5	Use Case Validation & Stakeholder Engagement	4	ANYWI	27.5	01	15
WP6	Connectivity & Communication Infrastructure	1	TUD	46	04	33
WP7	Unmanned Traffic Management & Coordination in U-Space	2	HHLAS	34	04	30
WP8	Operational Infrastructure	6	BYV	39	04	33
WP9	Public Acceptance Baseline & Use Case Oriented Survey	3	HFC	26.5	01	18
WP10	Experimental VR Study & Mitigation Strategy	3	HFC	23.5	13	33
WP11	Urban Drone Integration Strategy	5	TCN	40.5	04	33
WP12	Remote Piloting and HMI Interface	2	HHLAS	26.5	09	33
WP13	Emergency Response Piloting	6	IVD	51	09	33
WP14	Exploitation RP 1	7	IVD	24	01	18
WP 15	Exploitation RP 2	7	IVD	25	19	36
Total person months:				447		

Tables Quality and efficiency of the implementation #@QUA-LIT-QL@# #@WRK-PLA-WP@#-2: Work package descriptions

WP number	WP1	Lead beneficiary	TUD
WP title	**Project Management & Coordination RP 1**		

WP1 aims to ensure effective management of the project with emphasis on meeting the EC requirements. The specific objectives are:

- Ensure effective management of the project in line with EC requirements.
- Guarantee timely, high-quality research and innovation activities across all WPs.
- Implement an efficient management structure and decision-making platform.
- Monitor progress, risks, and deviations; apply mitigation measures.
- Facilitate transparent communication with the EC, partners, stakeholders, and the External Advisory Board (EAB).
- Ensure compliance with ethics, gender, and open access requirements.

Task 1.1 – Scientific and technical coordination (M01–M18; Lead: TUD; Others: all)

Objective: Coordinate scientific and technical activities across WPs and ensure achievement of objectives.

Activities: 1. The experienced H2020 and HEU coordinator, will coordinate activities and periodic technical reporting using an efficient management structure. 2. Set up and operate the Project Steering Board (PSB) (1 voting member per partner, chaired by coordinator). 3. Organize and chair kick-off and biannual meetings (incl. review meetings). 4. Monitor WP progress, risks, milestones, and deliverables. 5. Establish and manage the External Advisory Board for feedback and dissemination. 6. Supervise project implementation, conflict resolution, and knowledge transfer. 7. Ensure quality assurance and smooth inter-WP communication

Outcome: 1. Effective coordination and monitoring structure in place. 2. Transparent decision-making via Project Steering Board (PSB). 3. Timely delivery of milestones, deliverables, and periodic reports.

Task 1.2 – Administrative management (M01–M18; Lead: TUD; Others: all)

Objective: Ensure compliance with EC administrative, legal, and financial requirements.

Activities: 1. Manage legal, contractual, financial, and administrative issues. 2. Ensure compliance with Description of Action (DoA) and Consortium Agreement. 3. Collect, review, and submit deliverables, milestones, and reports to EC. 4. Budgeting, financial monitoring, and distribution of funds. 5. Facilitate partner–EC and partner–partner communication. 6. Maintain internal repository, website, and communication platform.

Outcome: 1. Timely and high-quality reporting to EC. 2. Transparent and efficient budget distribution. 3. Comprehensive repository for project documentation.

Task 1.3 –Gender , ethics, and data management (M01–M18; Lead: TUD; Others: all)

Objective: Ensure compliance with ethics, gender, GDPR and open-access requirements.

Activities:1. Draft, implement, and update the DMP. 2.Define data handling, collection, processing, and standards. 3. Select data protection and security measures.

Outcome: 1. Robust Data Management Plan (DMP) aligned with EC FAIR principles. 2. Ethics and gender issues monitored and addressed. 3. Open access ensured for all scientific data generated. 4.Monitor compliance with ethics, gender, and GDPR guidelines. 5.Update plans in response to new data, innovation potential, or consortium changes.

WP number	WP2	Lead beneficiary	TUD
WP title	**Project Management & Coordination RP 2**		

Objectives:

Identical to WP1 for second reporting period

Description of work:

Identical to WP1 for second reporting period. No critical deliverable – The outcomes of the WP focus on coordination and supporting demonstration, round-table stakeholder event and exploitation activities

WP number	WP3	Lead beneficiary	IVD
WP title	**Communication & Dissemination RP 1**		

Objectives:

The aim of this WP is to maximize the impact of the *dronnect* results through effective know-how transfer to a wide range of relevant audiences. This task is organized around the following key objectives:

- Prepare the dissemination and communication strategy, plan, channels and materials;
- Raise awareness, engagement and social acceptance among the key actors in the field and other appropriate audiences, including the general public;
- Produce actionable knowledge from the results to be disseminated through different tools and channels;
- Pave the way for market access of the *dronnect* solutions, technology, and processes and enhancing the exploitation routes;
- Develop a detailed exploitation plan covering all project results (individual and joint; commercial and non-commercial);
- Prepare business B2B models for viable and accelerated market evolution for drone logistics, and opportunities for *dronnect* solutions by defining their plans.

Description of work:

Task 3.1 – Project identity and online channels (M01–M18; Lead: IVD; Others: All)
Objective: Establish a strong, recognizable identity for the project and ensure online visibility at EU and international level.
Activities: 1. Design and deliver project logo, brochure/flyer, roll-up, and animated presentation video. 2. Develop, launch, and maintain project website (overview, results, publications, events). 3. Set up and operate social media channels like LinkedIn. 4. Leverage partners' social media to amplify outreach.

Outcome: 1. Professional project identity materials available. 2. Active and engaging online presence with broad visibility. 3. Stakeholder access point for project information and updates.

Task 3.2 – dissemination & communication strategy and implementation (M01–M18; Lead: IVD; Others: All)

Objective: Develop and execute an impact-driven multi-stakeholder communication and dissemination (C&D) strategy.

Activities: 1. Conduct stakeholder mapping and analysis. 2. Design C&D plan with tailored messages, channels, and tools. 3. Ensure open access to scientific publications and research data. 4. Monitor activities using indicators for outreach and engagement (online/offline). 5. Regularly update the plan based on impact assessment and gender-related considerations.

Outcome: 1. Clear, adaptive dissemination and communication strategy. 2.Strong engagement with stakeholders and wider community. 3. Increased visibility and impact of project outputs.

Task 3.3 – communication & dissemination Tools (M01–M18; Lead: IVD; Others: All)

Objective: Provide tools and channels to effectively communicate results and support uptake by stakeholders.

Activities: 1.Publish bi-annual eNewsletter. 2.Produce podcast series and video/digital storytelling content. 3.Organize periodic webinars and training sessions. 4.Publish best practice handbook. 5.Contribute to scientific publications and conference presentations.

Outcome: 1.Wide range of professional and engaging dissemination tools. 2.Enhanced stakeholder understanding and uptake of project achievements. 3.Contribution to knowledge sharing and capacity building.

WP number	WP4	Lead beneficiary	IVD
WP title	**Communication & Dissemination RP 2**		

Objectives:

Identical to WP3 for second reporting period

Description of work:

Identical to WP3 for second reporting period

WP number	WP5	Lead beneficiary	ANYWI
WP title	**Use Case Validation & Stakeholder Engagement**		

Objectives:

The aim of WP5 is to identify and address all potential risks associated with drone use-case pilots in VLL U-space airspace, with a particular focus on logistics applications. The following are key objectives

- Validate the use case and establish clear goals, expected outcomes, and implementation pipelines.
- Engage the stakeholders and ensure continuous interaction beyond scheduled meetings, fostering collaboration and knowledge exchange.
- Lay ground work for demonstrations by preparing specifications, risk assessments, and regulatory planning that will enable smooth execution of pilot projects.

Description of work:

Task 5.1 – Definition of use cases and specification requirements (M01–M06; Lead: ANYWI; Others: All)

Objective: Define detailed use cases for drone operations in logistics within VLL (Very Low Level) airspace.

Activities: 1. Prepare technical and operational requirements for drone-based logistics. 2.Identify potential risks and propose mitigation strategies. 3.Collect partner input to refine requirements and ensure cross-WP relevance.

Outcome: 1. Comprehensive set of use cases and requirements document. 2.Clear foundation for technology development, societal aspects, and governance WPs. 3.Risk-aware roadmap for subsequent implementation phases.

Task 5.2 – Initial stakeholder engagement survey (M04–M12; Lead: IVD; Others: All)

Objective: Gather early insights from key stakeholders to guide project direction and pilot design.
Activities: 1. Design and validate survey with industrial partners. 2.Engage regulators, city authorities, logistics companies, and technology providers. 3.Collect and analyse data on expectations, challenges, and acceptance of drone logistics. 4.Feed findings into preparation for Pilot 1 and 2, WP9/WP10 activities, and WP14 (business models).
Outcome: 1. Evidence-based understanding of stakeholder needs and concerns. 2.Early alignment of technical and societal developments with real-world requirements. 3.Stronger stakeholder involvement in shaping project pilots.

Task 5.3 – Pilot design and demonstration planning (M04–M15; Lead: ANYWI; Others: All)

Objective: Prepare detailed pilot designs and ensure readiness for demonstration activities.
Activities: 1. Develop pilot architecture and operational plans. 2.Identify certification requirements and compliance pathways. 3.Secure permissions and engage with relevant authorities in pilot cities. 4.Align with technical and societal work packages to ensure integration.
Outcome: 1. Approved pilot design and demonstration roadmap. 2.Permissions and certifications secured in advance. 3.Smooth transition to demonstration phase in Y3 Q3 with minimized regulatory and operational risks.

WP number	WP6	Lead beneficiary	TUD
WP title	**Connectivity and Communication Infrastructure**		

Objectives:

WP6 focuses on development and validation of technologies and algorithms to obtain reliable and resilient communication on critical links for safe drone operation. The aim is to create a trustworthy communications layer that will allow commercial operators to run their drone-based and mixed logistics systems without concerns about the availability and reliability of the network for remote control and piloting of the aircraft. By developing multi-link architectures (5G, satellite, mesh, Wi-Fi e-conspicuity), monitoring tools, handover schemes, and secure protocols (including post-quantum cryptography), WP6 provides the trustworthy communications layer needed for commercial operators and regulators to confidently scale drone logistics, in line with EASA initiatives objectives and U-space integration.

- Develop protocols and network management algorithms for resilient communication over commercial networks
- Develop QoS metrics suited for use in unmanned aviation to integrate with the network management systems
- Develop security algorithms suited for drone operations in U-Space and under SORA, including Post-Quantum Cryptography (PQC)
- Design and implement APIs and related architectures to integrate the communication link technologies into drone control software to enable BVLOS operation in commercial contexts
- Validate the technologies, algorithms and communication systems developed through test setups in logistics-relevant settings, such as port environments

Description of work:

Task 6.1: Design of link architecture (drone-several entities) and associated protocols(M04-M15; lead: TUD; others: ANYWI, HHLA, TCN, BYV)

Objective: Develop and integrate a real-time monitoring dashboard to track Command and control link performance, provide decision support for operators, and enhance safety through U-space integration.
Activities: 1.Define multi-link architecture including primary (5G), backup (satellite), and supplementary (mesh/Wi-Fi) channels. 2.Define link elements (ground link to remote operator, USSPs, Municipal authorities – contingency link). 3.Develop resilient network protocols for traffic prioritization, failover, redundancy, security algorithm, and prioritization of control messages. 4.Define QoS thresholds for C2, ensuring low latency, high reliability, and packet delivery guarantees. 5.Integrate e-conspicuity mechanisms to enhance situational awareness and compliance.

Outcomes: 1.Multi-link communication architecture for BVLOS operations. 2.Protocol specifications ensuring continuous and reliable command and control. 3.QoS metrics and operational requirements ready for testbed validation.

Task 6.2: Implementation and validation of the communication link testbed (M10-M21; lead: TUD; others: ANYWI)

Objective: Build and validate a multi-link testbed through simulated and hardware- real-world BVLOS trials, ensuring reliability, low latency, and robust failover strategies under operational conditions.
Activities: 1. Build a multi-link testbed supporting communication over 5G, satellite, mesh, and Wi-Fi. 2. Conduct simulated and real-world BVLOS tests to validate C2 link reliability, latency, and failover strategies. 3. Collect performance data, including availability, error rates, and QoS compliance. 4.Integrate testbed with drone control software for realistic command and monitoring scenarios.
Outcomes: 1. Operational C2 testbed validating multi-link communication. 2.Verified performance metrics and resilience under operational conditions especially for critical links. 3.Data for optimizing network management and safety-critical procedures.

Task 6.3: Implementation of network performance monitoring dashboard(M13-M24; lead: ANYWI; others: TUD, HHLAS, BYV,TCN)

Objective: Develop and integrate a real-time monitoring dashboard to track link performance, provide decision support for operators, and enhance safety through U-space integration.
Activities: 1. Develop dashboard to monitor link performance in real time, including link availability, latency, and failover status. 2.Provide decision support for remote operators, alerting on degradation or disruption of the especially C2 link. 3. Integrate monitoring with U-space platforms and operational control systems.
Outcomes: 1. Real-time network performance visualization tool for operators and traffic management. 2.Enhanced situational awareness and operational safety. 3.Feedback mechanism for continuous network optimisation with potential AI based algorithms.

Task 6.4: Evaluation of the testbed and resource management optimization (M25-M33; lead: ANYWI; others: TUD)

Objective: Evaluate testbed performance data to optimize network resource management, ensure scalability for multi-drone urban logistics, and prepare recommendations for integration into commercial BVLOS platforms.
Activities: 1.Analyse collected testbed data to evaluate link reliability and QoS compliance. 2.Implement network resource management and optimization algorithms to maximize link availability and efficiency. 3.Assess scalability to multi-drone and urban logistics corridors, including integration with U-space management systems.
Outcomes: 1.Validated communication system performance under operational conditions. 2.Optimized resource management algorithms for multi-drone operations. 3.Recommendations for integration into commercial BVLOS drone logistics platforms.

WP number	WP7	**Lead beneficiary**	HHLAS
WP title	**Unmanned Traffic Management & Coordination in U-Space**		

Objectives:

WP7 focuses on enabling safe, scalable, and harmonised BVLOS drone logistics operations within U-space and at its interfaces with ATM and non-U-space environments. The main objectives are:

- Analyse the role, requirements, and challenges of U-space for drone logistics.
- Address operational and technical challenges of BVLOS operations across U-space boundaries.
- Define coordination and communication mechanisms between USSPs, ATM, and UAS operators.
- Develop and validate workflows, solutions, and guidelines for seamless BVLOS integration.
- Assess U-space service implementation challenges and their alignment with ATM operations.

Description of work:

Task 7.1: Identify challenges of connecting U-space and non-U-space BVLOS logistics operations (M04-M09; lead: HHLAS)

Objective: 1. Identify operational challenges and technological gaps in connecting drone flights between U-Space and non-U-Space airspace, with additional input from stakeholders. 2.Define coordination, data, and communication requirements in these environments between USSPs, ATM and UAS-operators, to achieve strategic (and, where possible/ needed, tactical) deconfliction Activities: 1.Conduct a comprehensive review of current U-space regulations, standards (AMC/GM), and SESAR guidelines. 2.Map operational scenarios involving BVLOS flights entering/exiting U-space and identify risks, conflicts, and bottlenecks. 3.Identify technological gaps (e.g., communication protocols, interoperability of systems, data exchange requirements). 4.Engage stakeholders (air navigation service providers, regulators, drone operators, USSPs) through surveys, interviews, and workshops to gather practical requirements. 5.Define coordination, data-sharing, and communication requirements between USSPs, ATM, and UAS operators for deconfliction and safety-critical decision-making. Outcomes: 1.A clear definition of integration challenges when connecting U-space and non-U-space operations. 2.A requirements baseline for safe data exchange, coordination, and deconfliction mechanisms. 3.Stakeholder-informed gap analysis and operational needs to guide Task 7.2. 4.Identify operational challenges and technological gaps in connecting drone flights between U-Space and non-U-Space airspace, with additional input from stakeholders. 5.Define coordination, data, and communication requirements (WP6) in these environments between USSPs, ATM and UAS-operators, to achieve strategic (and, where possible/ needed, tactical) deconfliction. **Task 7.2: Develop workflows, solutions, and guidelines for seamlessly entering, exiting, and traversing U-spaces in BVLOS operations(M09-M24; lead: HHLAS; others: all)** Objective: 1.Design and develop safe and efficient operational and data workflows for realistic scenarios; validate with stakeholders. 2.Create solutions by adapting UTM- and UAS-control-center-platforms to share and manage flight intent, and evaluate their efficiency in both simulated and practical flight operations. 3.Draft guidelines, to support harmonization across Europe Activities: 1.Design safe, efficient, and interoperable operational workflows for BVLOS flights transitioning into, across, and out of U-space. 2.Develop data-sharing and communication workflows that link UTM platforms, UAS control centers, and ATM systems, ensuring interoperability and compliance with U-space rules. 3.Adapt and enhance UTM and UAS-control-center platforms to: a.Share/manage flight intent data. b.Support tactical conflict resolution where needed. c.Ensure efficiency in coordination with ATM and USSPs. 4.Validate workflows and solutions through simulation campaigns and selected practical test flights, in collaboration with stakeholders. 5.Draft European-level guidelines to harmonise practices across Member States and support standardisation. Outcomes: 1.Validated workflows and solutions for safe BVLOS integration into U-space Wp8. 2.Improved UTM/ATM platform interoperability with tested adaptations. 3.Draft guidelines to support harmonisation across Europe, aligned with SESAR and EASA activities. 4.Evidence base for scaling drone logistics operations across urban and peri-urban ecosystems.

WP number	WP8	Lead beneficiary	BYV
WP title	**Operational Infrastructure**		

Objectives:

This WP focuses on one of the most critical aspects of drone logistics operations in urban and peri-urban environments:

- Safe preparation, execution, and contingency handling of take-off and landing operations. With the growing reliance on U-space
- Possibility of Dynamic Airspace Reconfiguration (DAR), the project develops strategies, guidelines, and solutions for emergency and simultaneous multi-drone landings.
- WP8 underpins the demonstration activities in WP12 and WP13 by ensuring operational safety and compliance with EU aviation regulations.

Description of work:

Task 8.1: Identify challenges and strategy for preparation of take-off and landing sites(M04-M15; lead: BYV; others: HHLAS, TCN, ANYWI, TUD)

Objectives: 1.Define the operational, regulatory, and technical challenges for take-off and landing in dense urban and peri-urban environments. 2.Anticipate contingency situations (e.g., blocked landing pads, technical failures, weather constraints). 3.Determine adaptation measures beyond traditional GIS/map-based tools to ensure resilient landing site operations.
Activities: 1.Map operational challenges (urban density, infrastructure availability, public safety constraints). 2.Define requirements for site preparation, including safety margins, access routes, emergency services integration. 3.Develop contingency strategies for critical incidents during take-off and landing phases. 4.Identify digital tools (beyond static maps) for dynamic site allocation and risk assessment.
Outcomes: 1.A structured requirements baseline for safe and reliable drone operations in urban contexts. 2. Clear strategies for managing emergency take-offs and landings. 3.Identify challenges and key requirements about drone operations like take-off, landings in urban -peri urban scenario. 4.Solutions and strategy to handle critical contingency operations. 5.Need for adaptation measures above tools , suitable with AI based optimization for safe operations in urban-peri urban scenario.

Task 8.2: Analyse impact of disruptions in U-space through DAR issuance by ATM(M16-M24; lead: HHLAS; others: BYV, TUD; ANYWI, TCN)

Objectives: 1.Study the implications of DAR (Dynamic Airspace Reconfiguration) on drone operations. 2.Assess risks and impacts when large-scale disruptions force multiple UAS to land simultaneously. 3.Provide first-of-its-kind evidence on how DAR interacts with scaled drone logistics.
Activities: 1.Analyse EU regulations on DAR and their implications for urban drone ecosystems. 2.Simulate disruption scenarios (e.g., emergency airspace restrictions, sudden ATC reclamation). 3.Assess cascading effects on U-space services (e.g., communication, deconfliction, prioritisation). 3.Identify bottlenecks and propose resilience measures.
Outcomes: 1.Improved understanding of how DAR disrupts drone logistics in practice. 2.Strategies to minimise safety risks and operational breakdowns. 3.Evidence to inform EASA/SESAR standardisation and U-space safety-critical operations[21].

Task 8.3: DAR: Develop and test strategies, guidelines for simultaneous landing of multiple UAS(M24-M33; lead: HHLAS; others: BYV, TUD, ANYWI, TCN)

Objectives: 1.Develop guidelines for emergency landing scenarios involving multiple UAS. 2.Support logistics demonstrations in WP12 and WP13 by preparing operational procedures. 3.Ensure that DAR-triggered landings can be managed safely and efficiently.
Activities: 1.Design coordinated landing protocols for simultaneous landings in DAR situations. 2.Develop safety and contingency guidelines for emergency scenarios. 3.Validate procedures via simulation and feed into demonstration planning (WP12,13).
Outcomes: 1.First set of validated guidelines for handling simultaneous drone landings in U-space. 2. Increased resilience of U-space operations to emergency events. 3.Direct contribution to regulatory guidance for scalable drone logistics.

Advent to the drone logistics demonstration in WP 12, 13– provide initial guidelines and plan set of solutions to handle multiple landings, emergency landing, its effects in DAR and U-space integration

WP number	WP9	Lead beneficiary	HFC
WP title	**Public Acceptance Baseline & Use Case Oriented Survey**		

Objective

The objective of WP9 is to establish a robust and use case-specific understanding of public acceptance and societal readiness regarding drone operations in urban and peri-urban logistics contexts.

- Conduct a systematic literature review and representative public survey to assess societal attitudes, perceptions, fears, and concerns.

[21] U-space, as a functional safety critical system, includes the option to dynamically reconfigure airspace (in a CTR), as laid out in EU-regulations. In the case of scaled drone operations in such a U-space, multitude of drones will have to land in very short order to clear the airspace then “re-claimed” by ATC; the same can apply when a temporary airspace restriction is issued by authorities, e.g. in case of large scale emergencies. Such disruptions in UAS flight operations is not really understood, nor has it received adequate attention.

- Identify key acceptance factors such as perceived usefulness, operational transparency, and proximity to people and infrastructure.
- Provide empirical evidence to inform VR-based scenario studies, mitigation strategies, and broader drone deployment strategies.
- Ensure that public perspectives are meaningfully integrated into the design, planning, and operationalization of drone logistics systems.

Key focus:

- ➢ Identification of relevant societal acceptance factors through desk research
- ➢ Design and conduct of theory-based survey, including use case-specific aspects
- ➢ Quantitative analysis of public acceptance by use case
- ➢ Key insights for VR study and mitigations strategies (WP 10)

Description of work:

Task 9.1: Desk research on societal acceptance factors for drone logistics use case (M01-M06; lead: HFC*)*

Objective: Systematically identify societal acceptance factors relevant to drone operations, both general and logistics-specific.

Activities: 1.Conduct a comprehensive review of academic, industry, and regulatory literature on drone operations. 2.Identify societal concerns including safety, noise, privacy, and operational transparency. 3.Highlight logistics-specific factors such as visibility of operational purpose, proximity of drone activity to people and infrastructure, and perceived benefits. 4. Develop a conceptual framework for survey design in Task 9.2.

Outcome: Clear identification of key acceptance factors to guide survey development and refinement of drone acceptance models.

Task 9.2: Development of the use case-oriented survey (M05-M15; lead: HFC; others:all)

Objective: Design and implement a representative survey to assess societal readiness and public perceptions of drone logistics operations.

Activities: 1.Develop a comprehensive, theory-based questionnaire adapted to drone logistics use cases. 2. Integrate established constructs of public acceptance, including perceived usefulness, risk perception, and transparency. 3. Collaborate with a professional survey provider to define sampling rules, recruit participants, and ensure representativeness. 4.Conduct pre-tests of the questionnaire to validate clarity, reliability, and suitability. 5.Execute the survey to collect robust and representative data.

Outcome: 1.High-quality dataset capturing societal attitudes, fears, and acceptance drivers across different use cases. 2.Foundation for empirical analysis and input for VR studies and mitigation strategies.

Task 9.3: Empirical analysis of public acceptance by use case (M15-M18; lead: HFC)

Objective: Analyse survey data to evaluate societal acceptance and readiness for drone logistics operations.

Activities: 1.Perform quantitative and statistical analysis of survey responses. 2.Identify key drivers and barriers to acceptance, such as operational transparency, perceived usefulness, and proximity concerns. 3. Compare acceptance across different use cases to identify specific challenges and opportunities. 4.Provide actionable insights for demonstration planning, VR scenario development (WP10), and strategy formulation (WP11/WP12,13).

Outcome: 1.Detailed empirical understanding of public perceptions and societal readiness for drone logistics operations. 2.Identification of priority areas for mitigation and engagement strategies.

WP number	WP10	Lead beneficiary	HFC
WP title	**Experimental VR Study & Mitigation Strategy**		

Objectives:

In this WP, HFC’s existing drone acceptance model will be refined and expanded based on the findings of WP5 to include relevant use case-specific factors (logistics).

- Develop immersive Virtual Reality (VR) scenarios simulating representative drone operations in urban and peri-urban environments to evaluate public perceptions.

- Identify critical acceptance barriers and derive targeted mitigation strategies for successful integration of drone logistics operations.
- Compare VR-based acceptance data with real-world observations to assess transferability and reliability of findings. The key focus are:
  - Adaptation of HFC's drone acceptance model to (peri-)urban logistics use cases
  - Development and execution of immersive VR study based on project use cases
  - Derivation of mitigation strategies for deployment
  - Comparison of VR and real-world acceptance data to assess transferability

Description of work:

Task 10.1: Adaption of drone acceptance model to logistics use cases (M15-M20; lead: HFC)

Objective: Tailor the existing drone acceptance model to reflect logistics-specific operations in urban and peri-urban contexts.

Activities: 1.Integrate findings from the WP5,9 public survey to ensure societal attitudes, concerns, and expectations are captured. 2.Incorporate logistics-relevant acceptance factors, such as: visibility of operational purpose, transparency of drone activities , proximity to people, infrastructure, and sensitive areas 3.Identify potential barriers or triggers for public acceptance specific to logistics scenarios.

Outcome: 1. A refined drone acceptance model customized for urban and peri-urban logistics use cases. 2.Foundation for designing realistic VR-based evaluation scenarios.

Task 10.2: Development of Immersive VR environment based on Project Use Cases and Execution of VR study (M18-M33; lead: HFC; others: all)

Objective: Develop an immersive VR environment to simulate drone logistics operations and evaluate public acceptance in controlled, yet realistic, scenarios.

Activities: 1.Select up to three representative project use cases for the VR study based on stakeholder input.2. Design VR scenarios incorporating realistic variations, including: a. Drone flight altitude and path b. Operational environment (urban, peri-urban) c. drone size, appearance, and flight behaviour 3.Implement a modular VR setup to allow flexible simulation of different operational conditions. 4.Develop in-VR and post-VR questionnaires grounded in the adapted acceptance model (Task 10.1). 5.Recruit participants (n≥30) and conduct a pre-test to ensure validity and realism of scenarios. 6.Execute the VR study, collecting comprehensive feedback on perceptions, acceptance, and comfort levels.

Outcome: 1.High-quality VR-based data on public perceptions and acceptance of drone logistics operations. 2.Detailed understanding of scenario-specific factors influencing acceptance.

Task 10.3: VR-based study analysis and preparation of mitigation strategies (M24-M33; lead:HFC)

Objective: Analyze VR study results and develop strategies to mitigate public concerns and enhance acceptance of drone logistics operations.

Activities: 1.Conduct descriptive and exploratory analyses to identify acceptance levels and critical barriers across different scenarios. 2.Synthesize insights from WP5 public surveys and VR study findings. 3.Identify key issues affecting acceptance, including lack of transparency, unclear operational purpose, and proximity-related discomfort. 4.Develop practical recommendations and mitigation strategies for drone logistics operations.

Outcome: 1.Evidence-based mitigation strategies tailored to logistics-specific drone operations. 2.Guidelines for policymakers, operators, and urban planners to improve public trust and acceptance

*Results and findings will be made available to **WP15 (M36, lead: IVD)** for exploitation, ensuring that insights can inform future drone deployment in logistics and contribute to societal acceptance and mitigation strategy.*

WP number	WP11	Lead beneficiary	TCN
WP title	**Urban Drone Integration Strategy**		
Objectives:			

The aim of WP 11 is to

- Extend beyond conventional GIS-based urban planning by incorporating innovative drone logistics solutions into urban and peri-urban environments.
- Develop a digital platform to connect drone operators with municipal authorities, enabling real-time awareness and management of airspace within their jurisdiction.
- Facilitate smooth integration of drones into smart city frameworks and multi-modal urban mobility systems while addressing policy gaps, public acceptance, and regulatory requirements.

Description of work:

Task 11.1: Identification of policy gaps and model development of drone integration model for urban and peri-urban eco-system (M06-M15; lead: TCN; others: HHLAS, BYV, ANYWI)

Objective: Examine the current state of the art (SoA) and identify gaps in policies governing Very Low-Level (VLL) drone operations in urban and peri-urban areas.

Activities: 1.Comprehensive review of EU, national, and local policies and regulations for urban drone operations. 2.Identification of technical, operational, and legal barriers to integration of drone operations into urban ecosystems. 3.Definition of the functional and technical requirements of a digital platform to support interaction between municipal authorities and drone operators. 4.Development of a conceptual model for drone integration that aligns with urban planning, safety, and regulatory frameworks.

Outcome: 1.Clear understanding of policy gaps and operational constraints. 2.Blueprint for a digital platform that supports drone management in urban and peri-urban areas.

Task 11.2: Development and implementation of the application platform to involve municipal governance (M13-M29; lead: TCN; others: HHLAS, BYV, ANYWI, TUD)

Objective: Build and deploy a digital application platform to enable collaboration between drone operators and city authorities.

Activities: 1.Design and development of a user-friendly, secure, and scalable digital platform to visualize and manage urban airspace for drones. 2.Integration of functionalities for municipal authorities to monitor drone operations, issue permissions and restrictions, and receive notifications. 3.Demonstration of the platform with municipal stakeholders and collection of feedback on usability, functionality, and effectiveness. 4.Iterative refinement of the platform based on user feedback.

Outcome: 1.An operational platform connecting urban authorities to the U-space ecosystem. 2.Documented feedback and recommendations from city authorities to inform wider adoption.

Task 11.3: Analysis and strategy development for smooth integration in urban multi-modal transportation ecosystem (M25-M33; lead: TCN; others: HFC)

Objective: Develop strategies for the smooth integration of drones into urban mobility systems and smart city frameworks.

Activities: 1.Assessment of public perception and acceptance of drones in urban areas through stakeholder engagement. 2.Analysis of operational and regulatory challenges for multi-modal transport integration. 3.Development of strategic guidelines for integrating drone logistics with existing urban transport networks, considering safety, efficiency, and citizen engagement. 4.Communication of recommendations to municipal authorities to support policy development and operational implementation.

Outcome: 1.Guidelines and strategies for integrating drone logistics into urban mobility ecosystems. 2.Recommendations for policy and operational adjustments to support drone-enabled smart cities.

WP number	WP12	Lead beneficiary	HHLAS
WP title	**Remote Piloting and HMI Interface**		

Objectives:

The aim of WP12 is to achieve the mission of **pilot demonstration 1 as detailed in section 1**

• Conduct a public demonstrator of drone logistics operations to evaluate technical feasibility, operational performance, and on-site public acceptance. • Assess human-machine interface (HMI) requirements for remote operators managing multiple drones, supporting scalable and safe drone logistics operations. • Collect feedback from stakeholders, municipal authorities, and citizens to inform future deployment strategies and operational standards.
Description of work: **Task 12.1: Drone Logistics Pilot Demonstrator: preparation and final demonstration(M09-M24; lead: HHLAS; others: all)** Objective: Prepare, execute, and evaluate a public demonstrator of drone logistics in an urban or peri-urban environment. Activities: 1. Technical preparation of the drone network, including integration with remote operator platform and local communication systems. 2.Ensure compliance with regulatory requirements, including airspace permissions and safety protocols. 3.Develop communication and engagement strategies to inform the public and relevant stakeholders about the demonstration. 4.Invite reviewers, city authorities, and stakeholders to observe the demonstration. 6.Execute drone logistics operations with real-time monitoring through the remote operator platform. 7.Provide municipal authorities access to the dashboard for situational awareness and monitoring of network performance. 8. Conduct an on-site public acceptance study to gather perceptions, concerns, and satisfaction from attendees. Outcome: 1.Successful demonstration of drone logistics operations in a realistic urban environment. 2.Evaluation of technical performance, operational workflow, and stakeholder engagement. 3.Collection of data on public perception and acceptance of drones in urban and peri-urban environment. **Task 12.2: Feasibility study of HMI interface (one remote pilot to multiple drones) (M18-M24; lead: HHLAS; others: BYV, ANYWI, TUD, HFC)** Objective: Evaluate the capabilities and requirements for remote operators managing multiple drones simultaneously. Activities: 1.Analyse lessons learned from the demonstrator to define operational requirements of HMI interfaces. 2.Assess skills, training, and cognitive load needed for remote operators. 3.Conduct a realistic study of multi-drone fleet management, including simultaneous communication with drones and coordination with municipal authorities and multi-level governance structures. 4.Identify challenges and limitations in remote monitoring, situational awareness, and decision-making. Outcome: 1. Detailed requirements and recommendations for HMI design to enable efficient and safe operation of multiple drones. 2.Guidelines for training and skills needed by remote pilots managing drone fleets in urban logistics operations.

WP number	WP13	Lead beneficiary	BYV
WP title	**Emergency Response Piloting**		
Objectives: The WP 13 aims to accomplish the goals of **pilot demonstration 2 aforementioned in section 1** Address the lack of knowledge on procedures for emergency drone operations in urban environments. • Evaluate public understanding and interaction with emergency drone landing. • Coordinate with municipal authorities via their information platforms to ensure proper procedures and oversight. The following are key focus: ➢ Preparation, execution, and evaluation of an emergency response demonstrator using drones. ➢ Assessment of societal acceptance and understanding of emergency drone procedures. ➢ Derivation of lessons learned and recommendations for systematic integration of emergency drone operations in urban areas.			
Description of work:			

Task 13.1: Emergency response Demonstrator: preparation and final demonstration(M15-M33; lead: BYV; others: TCN, ANYWI, HHLAS, TUD, HFC)

Objective: Prepare and execute a pilot demonstration of emergency response drones, involving municipal authorities and the general public.

Activities: 1.Obtain necessary certifications, licenses, and permissions for urban emergency drone operations. 2.Coordinate with municipal authorities to ensure integration with local emergency information platforms. 3.Inform and involve the public in preparation for the demonstration. 4.Execute the demonstration, showcasing emergency drone deployment for medical or critical incident response. 5.Conduct on-site public acceptance study to evaluate understanding, comfort, and confidence in drone-based emergency procedures.

Outcome: 1.Successful demonstration of emergency drone operations in coordination with city authorities. 2. Data on public perception and acceptance of emergency drones in real urban contexts.

Task 13.2: Analysis and lessons learnt and strategies to systemize the operation(M28-M33; lead: BYV; others: IVD, HFC)

Objective: Analyse the results of the emergency response demonstrator and develop strategies for standardized, repeatable operations.

Activities: 1.Identify lessons learned regarding technical execution, operational coordination, and public engagement. 2 Compare on-site public acceptance results with VR-based acceptance data from WP10 to evaluate transferability and scenario validity. 3.Document procedural requirements, societal considerations, and operational recommendations. 4.Prepare an observation and experience report to be communicated via the project's Exploitation WP for broader dissemination and uptake.

Outcome: 1.Consolidated insights on emergency drone operations in urban environments. 2.Guidelines and strategies to systemize emergency drone response for future deployment.

WP number	WP14	Lead beneficiary	IVD
WP title	**Communication & Dissemination RP 1**		

Objectives:

This WP ensures that the results of *dronnect* contribute to the fast-paced market deployment of drone logistics while aligning with EU aviation regulations, traffic management standardisation (U-space, SESAR, EASA AMC/GM), and societal acceptance.

- It evaluated both technical and non-technical exploitation pathways, including business models, roadmaps to standardisation, and societal impact mitigation strategies.
- Develop a comprehensive exploitation plan covering all project results, addressing both commercial and non-commercial opportunities.

Description of work:

Task 14.1 – Exploitation Plan and Implementation (M01–M18; Lead: IVD; Others: All)

Objective: Define and implement a comprehensive exploitation plan ensuring uptake and sustainability of project results. Activities: 1.Identify and list exploitable results (commercial and non-commercial) across all WPs. 2.Specify result types, IP ownership, access rules, marketability, and exploitation pathways. 3.Agree on IP protection measures considering background and foreground knowledge. 4.Conduct market analysis via desk research and stakeholder interviews/surveys. 5.Collect partner input during two dedicated exploitation workshops. 6.Finalize exploitation plan towards end of project to support post-project uptake.

Outcome: 1.Consolidated list of exploitable results with clear ownership and access rules. 2.Joint and individual exploitation plans validated by partners. 3.Market analysis report supporting result commercialization and adoption. 4.Final exploitation plan enabling sustainability beyond project duration.

Task 14.2 – Stakeholder Study and B2B Drone Logistics Business Model (M01–M24; Lead: IVD; Others: ANYWI, HHLAS, TCN, BYV, external stakeholders)

Objective: Develop a robust B2B drone logistics business model aligned with stakeholder needs and market dynamics. Activities: 1.Conduct internal stakeholder survey (logistics, drone operators, authorities, end-users). 2.Engage external stakeholders to capture market expectations and operational needs. 3.Assess stakeholder

feedback after first public demonstration and integrate into model. 4.Analyse economic, operational, and regulatory feasibility of drone logistics. 5.Draft and validate business model framework in collaboration with industrial partners. Outcome: 1.Evidence-based stakeholder study informing project pilots and business case. 2.B2B business model for drone logistics grounded in stakeholder demand and demonstration insights. 3.Clear pathways for commercialization and scale-up of project outcomes.

WP number	WP15	Lead beneficiary	IVD
WP title	**Exploitation RP 2**		

Objectives: The final WP of the project aims to bring the knowledge, and solutions of *dronnect* that paves a way towards sustainable actions after successful completion of the project

- It delivers both technical and non-technical exploitation pathways, including business models, roadmaps to standardisation, and societal impact mitigation strategies.
- Prepare for market launch of solutions by defining suitable B2B business models, deployment strategies, and fast-track operational plans.
- Analyse societal acceptance outcomes and provide mitigation strategies to enhance public trust and facilitate adoption of drone logistics solutions.

Description of work:

Task 15.1: Roadmap towards communication infrastructure and traffic management system standardization (M28-M36; lead: ANYWI; others: TUD, TCN, HHLAS, BYV)

Objectives: 1.Capitalises on lessons learnt from demonstrations and technical pilots. 2.Align project results with EU-level traffic management initiatives (EASA AMC/GM, SESAR, U-space). 3. Provide clear technical and regulatory guidelines for safe and scalable integration of drones.
Activities: 1. Consolidate lessons learnt from technical pilots (WP6–13). 2.Evaluate system designs and solutions, focusing on safety-critical communication infrastructure. 3.Draft guidelines and recommendations for EU and national regulators. 4.Engage with standardisation bodies and policy stakeholders to validate outputs. Outcomes: 1.A roadmap to standardisation supporting U-space services and EU drone strategy. 2.Evidence-based guidance to EASA, SESAR, and other bodies for regulation updates. 3.Higher certainty for industry and regulators on scalable drone integration.

Task 15.2: Analysis of suitable model and strategies and tools for fast paced deployment (M25-M33; lead: IVD)

Objectives: 1.Identify the most effective B2B business models for drone logistics. 2.Develop strategies and tools for rapid market deployment in urban and peri-urban contexts.
Activities: 1.Analyse lessons from stakeholder engagement (WP12) and demonstrations (WP 12,13) and WP14. 2. Design, test, and validate a B2B business model tailored to logistics operators. 3.Map pathways for accelerated adoption in different EU city contexts.
Outcomes: 1.A validated B2B business model aligned with EU market dynamics. 2.Practical roadmap for fast-paced deployment of drone logistics services. 3.Increased certainty for investors and operators in urban mobility ecosystems.

Task 15.3: Public acceptance and social impact and mitigation strategies for drone logistics use case (M28-M33; lead: HFC)

Objectives: 1. Assess public acceptance of drone logistics through real-life and immersive scenarios. 2.Analyse societal concerns and provide mitigation strategies. 3.Translate findings into actionable recommendations for industry, municipalities, and regulators.
Activities: 1.Synthesize results from public surveys, VR-based experiments, and on-site demonstrations. 2.Analyse key social impacts (privacy, safety perception, noise, trust). 3.Develop evidence-based mitigation strategies (communication, transparency, governance).
Outcomes: 1.Clear understanding of societal concerns and expectations regarding drone logistics. 2.Mitigation strategies to reduce social tensions and build trust in the technology. 3.Strong contribution to EU missions on climate-neutral and smart cities by ensuring societal alignment.

Table Quality and efficiency of the implementation #@QUA-LIT-QL@# #@WRK-PLA-WP@# 3: List of deliverables

Del. No.	Deliverable name	WP No.	Lead Partici-pant	Type	Dissm. level	Due date (in months)
D1.1	Project handbook and Update	WP 1	TUD	R	EU-C	M3
D1.2	Data Management Plan and Updates			R	EU-C	M6
D1.3	Report on Ethics and Gender and Update			R	EU-C	M18
D3.1	Project Website	WP 3	IVD	DEC	PU	M3
D3.2	Communication Kit			R	PU	M3
D3.3	Plan for dissemination and exploitation of results and communication activities			R	SEN	M6
D3.4	Report on mid-term communication and dissemination activities			R	PU	M18
D4.1	Final report on dissemination and communication activities	WP 4		R	PU	M36
D5.1	Use-case Definition and Requirements for drone BVLOS operation	WP 5	ANYWI	R	EU-C	M6
D6.1	Requirements of communication infrastructure and link architecture for drone BVLOS operation	WP 6	TUD	R	EU-C	M18
D7.1	System Requirements Specification Report	WP 7	HHLAS	R	SEN	M9
D7.2	Platform Implementation Report		HHLAS	R	SEN	M30
D8.1	Operation infrastructure and safety requirements in urban, peri-urban environment	WP 8	BYV	R	EU-C	M9
D8.2	Impact Analysis Report		HHLAS	R	SEN	M24
D9.1	First report on social readiness and survey model	WP 9	HFC	R	EU-C	M6
D9.2	Summary report of public-acceptance assessment and findings			R	EU-C	M18
D11.1	Strategy and solution towards integration of city and municipal authorities in enabling drone operation in urban and peri-urban ecosystem	WP 11	TCN	R	EU-C	M33
D12.1	Round-table event to engage with and gather feedback from stakeholders/ potential users for drone logistics	WP12	HHLAS	DEM	PU	M18
D12.2	“Open day” with demo of drone logistic operation for interested users/ public, round-table discussion		HHLAS	DEM	PU	M33
D13.1	Pilot demonstration for an emergency response coordinated with city authority’s platform	WP 13	BYV	DEM	PU	M33
D14.1	Exploitation Plan and Implementation Updates	WP14	IVD	R	EU-C	M18
D15.1	HMI considerations for operators controlling multiple drones simultaneously	WP 15	IVD	R	EU-C	M24
D15.2	Summary of recommended procedures and guidelines to minimize disruptions by DAR			R	EU-C	M33
D15.3	Final report on societal acceptance and mitigation strategies			R	EU-C	M36
D15.4	Roadmap to standardization of traffic management system, safety-critical operational and communication infrastruc-ture			R	EU-C	M36
D15.5	Roadmap for fast deployment of B2B drone operations in urban and peri urban ecosystem			R	EU-C	M36

Table Quality and efficiency of the implementation #@QUA-LIT-QL@# #@WRK-PLA-WP@#-3: List of milestones

No.	Milestone name	Related WPs	Due date	Means of verification (Deliverables)
M01	Project Launch and Foundation	WP 1, WP3	M6	D1.1, D1.2, D3.1, D3.2
M02	Requirements and Baseline Definition	WP 3, WP5,	M6	D3.3, D5.1
M03	Technical Architecture and System Design Finalization	WP 6, WP7, WP 8, WP9, WP11	M18	D6.1, D6.2, D7.1,D8.1,D8.2, D9.1,D9.2
M04	Pilot 1: dronnect demo day & Stakeholder Engagement	WP 12	M24	D12.1, D12.2
M05	Pilot 2: Emergency response & Local Authorities Integration	WP 13	M33	D13.1
M06	Project Completion and Exploitation Roadmap	WP 10, WP 14, WP 15	M36	D14.1, D14.2, D15.1, D15.2, D15.3, D15.4, D15.5

Table Quality and efficiency of the implementation #@QUA-LIT-QL@# #@WRK-PLA-WP@#-4: Critical risks for implementation #@RSK-MGT-RM@#

Risk	Related WPs	Risk Description	Impact	Mitigation Measures
R01	WP12, WP13	[Technical] Regulatory delays and evolving U-Space requirements	High	Early engagement with authorities for reviewing certification, licensing, and approval processes directly after WP5. Allocate sufficient time for Demonstration (M33).
R02	WP9, WP10	[Technical] Public acceptance lower than expected	Low	Use preliminary WP9 results to adapt WP10 activities. Conduct public surveys during demonstrations to gather feedback and address concerns promptly.
R03	WP9, WP12	[Societal] Insufficient public response	Medium	Through WP 3,4 gain attention to the public survey. Post D9.1 effectively perform a large scale public outreach. Leverage the outcomes of WP5 to prepare for public demonstration and survey in WP 12(M33).
R04	WP6, WP7, WP8, WP11	[Technical] Technology Readiness Level (TRL) not achieved	High	Implement staged TRL progression with internal validation gates and go/no-go decision points. Explore alternative technical approaches for critical components in early research tasks.
R05	WP11, WP12, WP13	[Technical/Administrative] Municipal authority engagement difficulties	Medium	Provide clear value propositions for municipalities. Develop flexible engagement models (Task 12.1) to accommodate varying municipal structures.
R06	WP1, WP2	[Administrative] Consortium partner coordination challenges	Low	Apply robust project management with monthly reviews. Establish clear WP interdependency protocols. Use shared technical platforms (as in D1.2) for real-time collaboration.
R07	Cross-cutting	[Financial/Technical] System requirements or integration misalignment	Medium	Conduct iterative requirements gathering with stakeholder validation. Define early integration protocols, perform continuous integration testing, and ensure open communication through frequent online meetings and shared workspaces.

Table5: Summary of staff effort

Staff effort	WP1	WP2	WP3	WP4	WP5	WP6	WP7	WP8	WP9	WP 10	WP 11	WP 12	WP 13	WP 14	WP 15	Total
1-TUD	8	10	2	2	3	15	4	3	1	1	5	3	3	1	1	62
2-HHLAS	1	1	1.5	1.5	1.5	5	18	14	1	0.5	5	12	6	4	4	76
3-HFC	1.5	1.5	3	3	3	0	1	0	18	16.5	1	5	4.5	2	3	63
4-ANYWI	1	1	0.5	0.5	12	12	4	6	2	3	6	3	2	3	3	59
5-TCN	1.5	1.5	1.5	2	2	5	2	2	1	1	18	1	9	1	1	49.5
6-BYV	1.5	1.5	3	3	4	9	4	14	1.5	0.5	4.5	2	26	2	2	78.5
7-IVD	2	3	12	12	2	0	1	0	2	1	0	0.5	0.5	11	11	58
Total	16.5	19.5	23.5	24	27.5	46	34	39	26.5	23.5	40.5	26.5	51	24	25	447

Table6: 'Subcontracting costs' items

Subcontracting	Costs (€)	Description of tasks and justification
2/HHLAS	400,000	Task 6.1: Involvement of external (subcontracted) dev resources needed to ensure a good fit of the proposed pathway to HHLA Sky's platforms Task 6.2: Code adaptations and iterations needed in comms pipeline and ICC logic, to allow safe, secure, and reliable communications drone <-> backend (Coding is outsourced, as is customary). Task 6.3: Integration of key features of dashboard into operator tools (ICC) and programming of API, to monitor comms states Task 6.4: Iterative testing and software troubleshooting to ensure suitability and performance Task 7.2: Adaptation of traffic management system (code adaptation and development and provision of needed APIs) to support the developed workflows and integration with ATC Task 8.3: Adaptation of traffic management system (code adaptation and development and provision of needed APIs) to support developed strategies and enable a prototype integration with external systems (municipality systems)

Table 7: 'Purchase costs' items

	Costs (€)	Justification
2/HHLAS		
Travel and subsistence	23,000	Travel to consortium & review meetings (10k€), travel to workshops and fairs/conferences (7k€), travel to partners, for testing/demo (6k€)
Equipment	277,000	Depreciation costs for 2 V25 eVTOL (2x63k€); for an X25 Multicopter (55k€) and costs for Integrated Control Center & UTM (license fees, setup, hosting and maintenance (96k€)
Total	**300,000**	
6/BYV		
Travel and subsistence	25,000	Travel to consortium & review meetings (20k€), travel to workshops and fairs/conferences (5k€)
Equipment	36,500	Depreciation costs for server (15k€), inertial measurement unit (10k€), IR sensor (5k€), RGB sensor (2k€), high-precision RTK (2k€), laptop (2.5k€)
Other goods, works and services	18,500	Electronics and mechanics consumables and high performance battery (6.5k€), dissemination material and fees for fairs/conferences (5k€),
Total	**80,000**	

Table8: 'Other costs categories' items

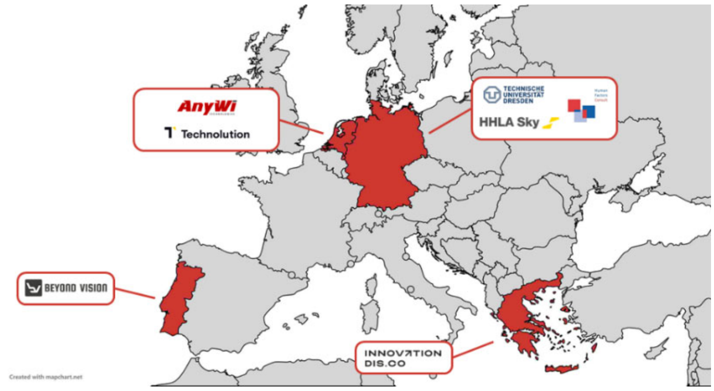
	Costs (€)	Justification
2/HHLAS		
Internally invoiced goods and services	200,000	adaptation of internal flight control systems/ components and logic workflows
No/Partner short name		

Table 9: 'In-kind contributions' provided by third parties – NOT APPLICABLE #§QUA-LIT-QL§# #§WRK-PLA-WP§#

CAPACITY OF PARTICIPANTS AND CONSORTIUM AS A WHOLE #@CON-SOR-CS@# #@PRJ-MGT-PM@#

dronnect consortium consists of **7 partners from 4 European countries**, combining complementary expertise from research, industry, public, local authorities, technology providers. Geographic and institutional diversity is illustrated in **figure5**, reflecting project's commitment to cross-border collaboration relevance. The consortium has been carefully formed to ensure full coverage of all scientific, technical, societal aspects required to address the project's objectives. Partners were selected based on their domain-specific expertise, successful track record in collaborative R&D, ability to contribute - innovation and implementation.

Figure 3: IAM - countries covered and partners involved



TUD academic excellence and cross-domain research capacity is led through the ComNets Chair with projects e.g, CeTI, 6G-Life, AirMetro, research, connection with global expertise in robotics, HMI, advanced communication architectures. The research focus positioned at the forefront of 6G, AI-driven communication systems to aerial devices via **terrestrial and, Non-Terrestrial Network (NTN) based solutions, which will underpin the multi-link communication**, tactile internet for remote operation aspects of the project. TUD has a strong track record in management, coordination of Horizon projects with specialist in the department of **European Project Centre**. **HHLAS**, a German based industry leader in providing **modern, service-based technology platforms to support scalable UAS and mobile robots operations**. Their **Integrated Control Centre (ICC)** is capable of simultaneously managing more than 100 drones, mobile robots securely and efficiently, and is cyber-security certified by **TÜV Nord acc. to IEC 62443** – still a world's first in this realm. It has aviation-based standard operating procedures built-in to support safe BVLOS missions for commercial and government users. **HHLAS's UTM platform** benefits from 5 years of dedicated research in the field, has a strong focus on integration and compliance to **EU regulation 2021/664** and is certifiable. The platforms can intelligently integrate with data e.g. from ERP-solutions, external supporting services and many other systems, such as ships, air traffic information, governmental geozone providers, make data available to external stakeholders. This ensures operational realism, transferability into sustainable urban drone logistics. **HFC**, a German applied **research institute specialized in human-machine interaction, usability, and public acceptance of emerging technologies.** It brings **extensive experience from EU projects on drone systems, including large-scale surveys and stakeholder research** (e.g. ADACORSA). HFC leads the work on societal readiness, public perception, acceptance strategies using survey,VR-based experimental methodologies. **ANYWI** is a Dutch industry specializing in **communication software architectures and secure connectivity solutions.** ANYWI develops robust IP communication systems optimized for reliability, multipath redundancy, with proven experience in wireless networks embedded software. Their expertise directly supports the project's goal of developing resilient multi-link communication architectures for BVLOS operations. **TCN**, a Dutch company with leading expertise **in traffic management software, multimodal mobility platforms.** Their **flagship product, MobiMaestro**, integrates real-time transport mobility data, enabling cities to dynamically manage traffic, policy objectives, infrastructure usage. TCN will transfer its proven large-scale traffic management know-how into the aviation domain, ensuring seamless interaction between drone U-space services, wider urban mobility ecosystems. **BYV** is a Portuguese **UAV manufacturer and operator, specialized in autonomous drones and UTM compatible mission platforms.** They design AI-powered UAV systems, including the **beXStream cloud management solution,** enabling autonomous BVLOS operations with advanced automation, obstacle avoidance, fleet management functionalities. BYV has delivered UAV fleets to the Portuguese Navy, demonstrating maturity in certified UAV manufacturing, operational deployment, integration with safety-critical users. **IVD**, a Greek SME **focusing on innovation management, communication, societal integration of R&I projects.** With more than 20 years of experience in dissemination, stakeholder engagement across European mobility programmes, they will ensure societal acceptance, visibility, exploitation of results. IVD strengthens the project's impact dimension by actively connecting with municipalities, UAM networks, policy stakeholders. #§CON-SOR-CS§# #§PRJ-MGT-PM§#